

**VIETNAM GENERAL CONFEDERATION OF LABOR
TON DUC THANG UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY**



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COMPUTER SHOP WEBSITE

FINAL REPORT REQUIREMENT ANALYSIS AND DESIGN

**Instructor
Ms. PHAN THI THANH PHUONG**

HO CHI MINH CITY, 2025

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Ho Chi Minh, May 19, 2025

Author

(Ký tên và ghi rõ họ tên)

DECLARATION OF AUTHORSHIP

Our team would like to pledge that this is our own research project and is guided by teacher Phan Thi Thanh Phuong. The research content and results in this topic are honest and have not been published in any form. any before. The data in the tables for analysis, comments, and evaluation were collected by the author from different sources and clearly stated in the reference section.

In addition, the project also uses a number of comments, assessments as well as data from other authors and other organizations, all with citations and source notes.

If any fraud is detected, our team will take full responsibility for the content of their essay report. Ton Duc Thang University is not involved in copyright violations caused by us during the implementation process (if any).

Ho Chi Minh City, May 18, 2025

Authors

COMPUTER SHOP WEBSITE

ABSTRACT

The project presents the development of a computer shop website that allows employees to manage the shop online, and enables customers to browse, search and purchase products online. The system includes key features such as account registration, product management, customer management, order creation, and employee management. The website is built using modern technologies like React for the frontend, Express.js for the backend, and Mysql as a database management.

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CHAPTER 1. INTRODUCTION

1.1 Overview

The Computer shop website is an e-commerce platform designed to provide a smooth online shopping experience for customers looking for an appropriate device for them. The system enables users to browse products by many methods, such as searching by name, filtered items by specifications (e.g., brand, RAM, storage,...).

1.2 Objectives

- Offer customers a user-friendly UI to explore and purchase products.
- Allow admin to efficiently manage employees, customers, products,...
- Provide secure authentication for both employees and customers.

1.3 Target

- Customers: Browse, filter, and purchase products.
- Admin: Manage inventory, customers, employees, orders,...
- Salesperson: Manage inventory, customers, orders,...
- Accountant: Manage orders.

1.4 Technologies used

- Frontend: React JS
- Backend: Node.js, Express.js
- Database: MySQL
- Styling: TailwindCSS
- Authentication: JWT Auth

1.5 Development model used

This project used the **Agile development model**, which divides the work into small parts called sprints. Each sprint focuses on building specific features. Agile allows for easy changes, quick feedback, and continuous

improvement. This made the development process more flexible and helped create a better and more user-friendly website.

CHAPTER 2. REQUIREMENT DOCUMENTATION

2.1 Functional Requirement

- **User login:** Users are allowed to login to their account to see necessary information like profile, orders, products, sale history,...
- **Manage product:** Salespeople can view inventory, product details, and quantity. Admin can add new products, update products and delete products. The product quantity will automatically be decreased according to the quantity bought by customers.
- **Manage staff:** For admin only. Admin can add new staff, delete staff, lock/unlock staff, send verification email when staff forget their password.
- **Manage sale history:** Accountant is responsible for managing sales per month, including revenue, profit, export invoices, view invoice details.
- **Manage personal info:** Users can change avatar, name, email.
- **Manage orders:** Employees, admin can view, add, update, and delete orders, while customers can only view + place orders.
- **Manage personal information:** All users can view, update their information.

2.2 Non-functional Requirement

- **Performance:**
 - + System shall be able to handle up to 1000 users at a time without any problems.
 - + The response time shall be within 3 seconds.
- **Scalability:**

- + The system shall be designed to store up to 1000 products and support a growing number of customers.
- **Portability:**
 - + The system shall work on different browsers such as Chrome, Edge, Firefox,...
- **Reliability:**
 - + The system shall have uptime of at least 99.5% every month.
 - + In case of a system crash, it shall recover within 5 minutes.
- **Security:**
 - + Users shall login to access the system, and the password is encrypted.
 - + The system shall use role-based access control, meaning that only authorized users can perform the specific actions.
- **Usability:**
 - + The UI shall be user-friendly and easy to use.
 - + All the error messages shall be clear.

CHAPTER 3. IDENTIFY ACTORS AND USE CASES

3.1 Actors

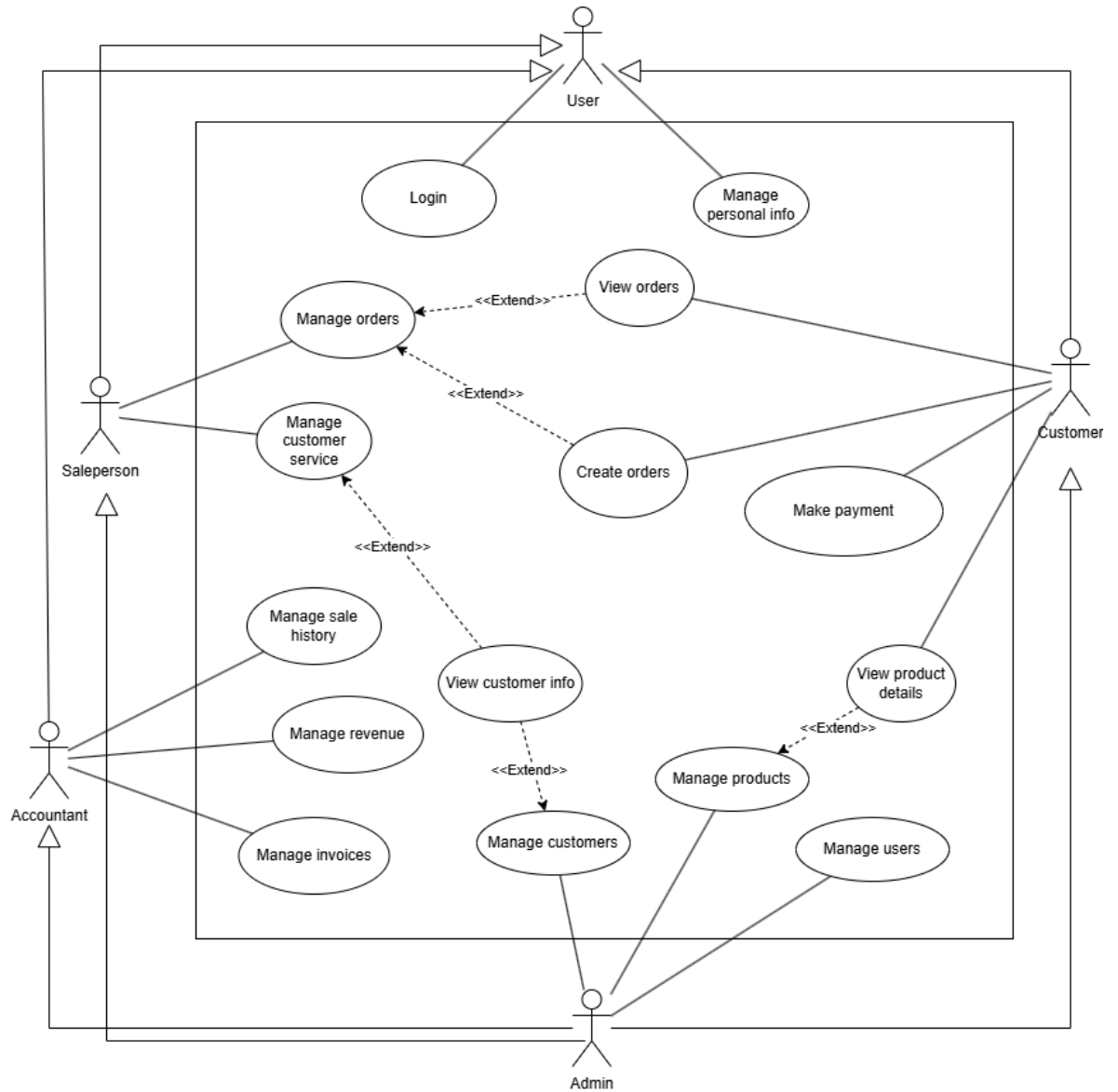
- Admin: Has full control over the system.
- Salesperson: Can manage customers, products, orders, manage personal info.
- Accountant: Can manage orders, personal info.
- Customer: Can view products, orders, place orders, manage personal info.

3.2 Use cases list

Use case ID	Use case name
UC01	Login
UC02	Manage personal info
UC03	Manage orders
UC04	Manage customer service
UC05	Manage sale history
UC06	Manage revenue
UC07	Manage invoices
UC08	Manage customers
UC09	Manage products
UC10	Manage users
UC11	Make payment

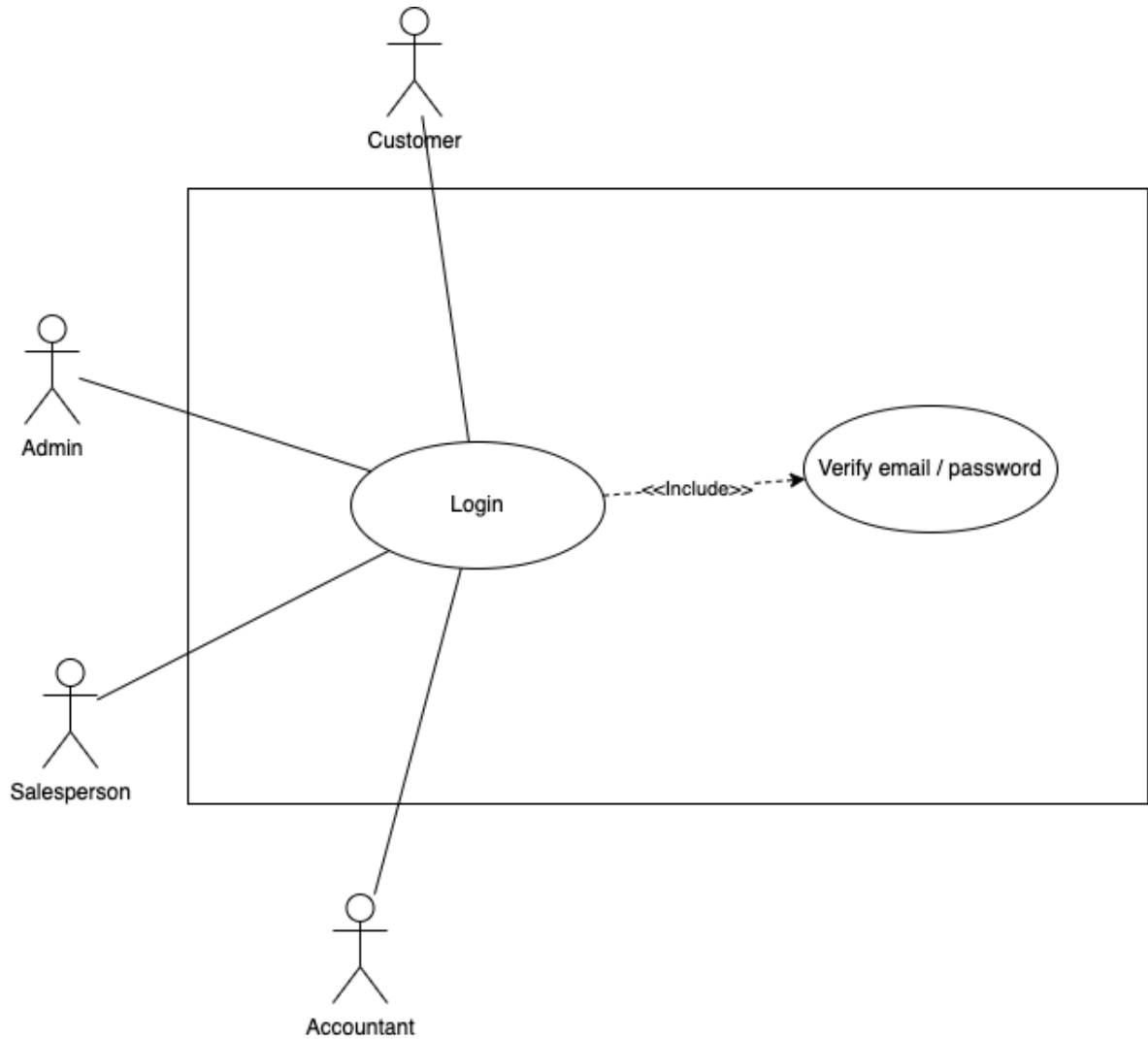
CHAPTER 4. USE CASE DIAGRAM

4.1 General Use Case Diagram

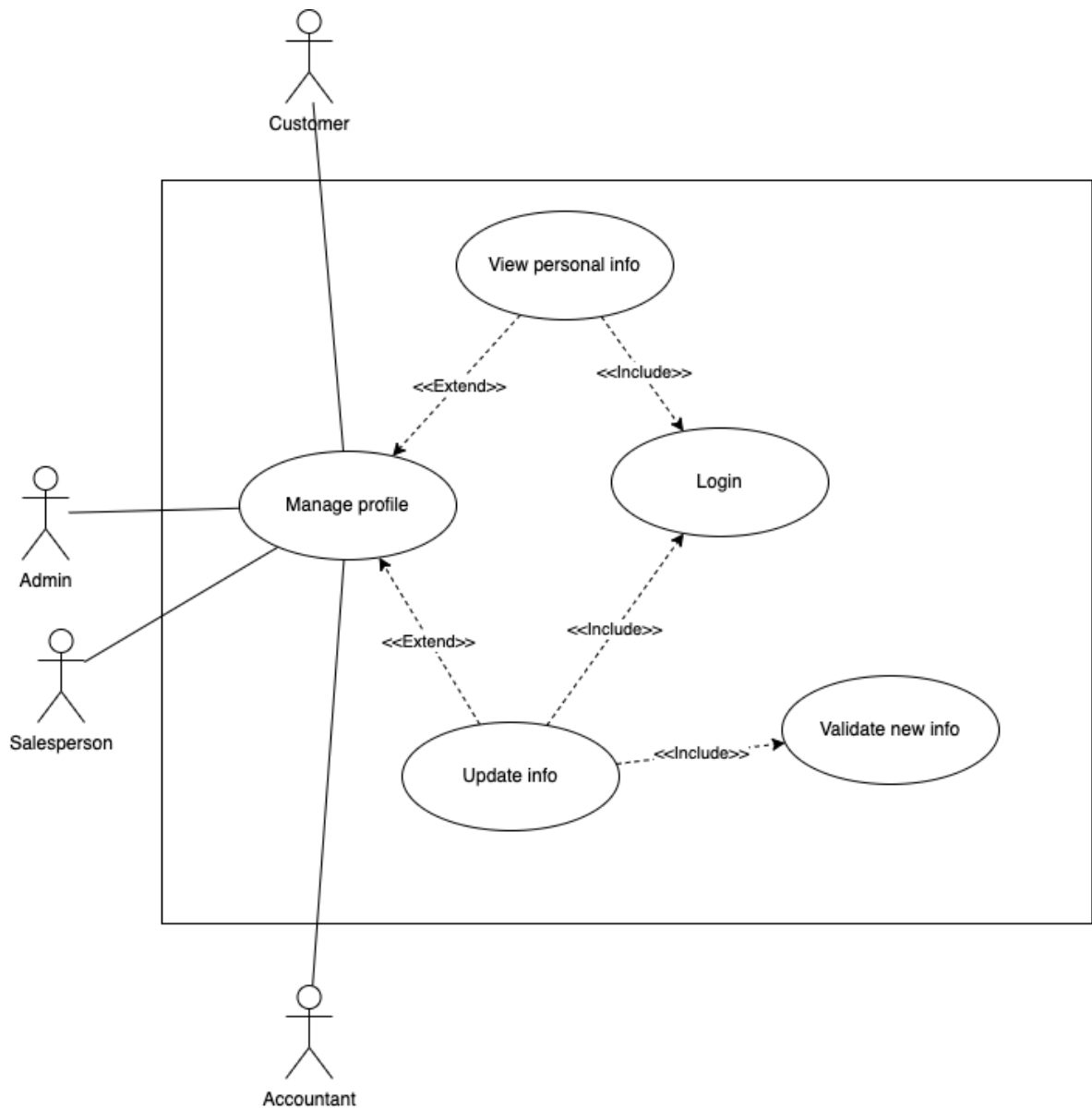


4.2 Detailed Use Case Diagram

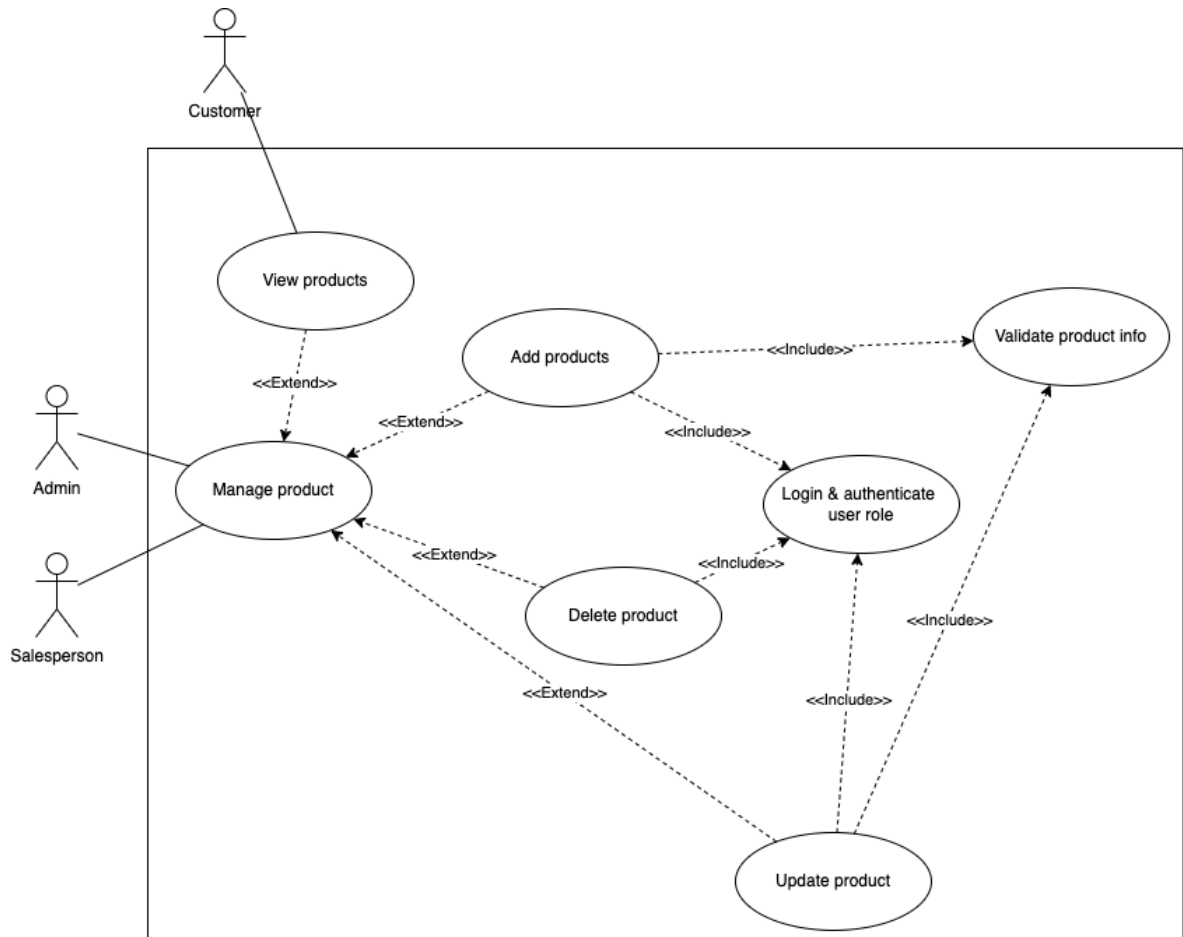
4.2.1 Login use case diagram



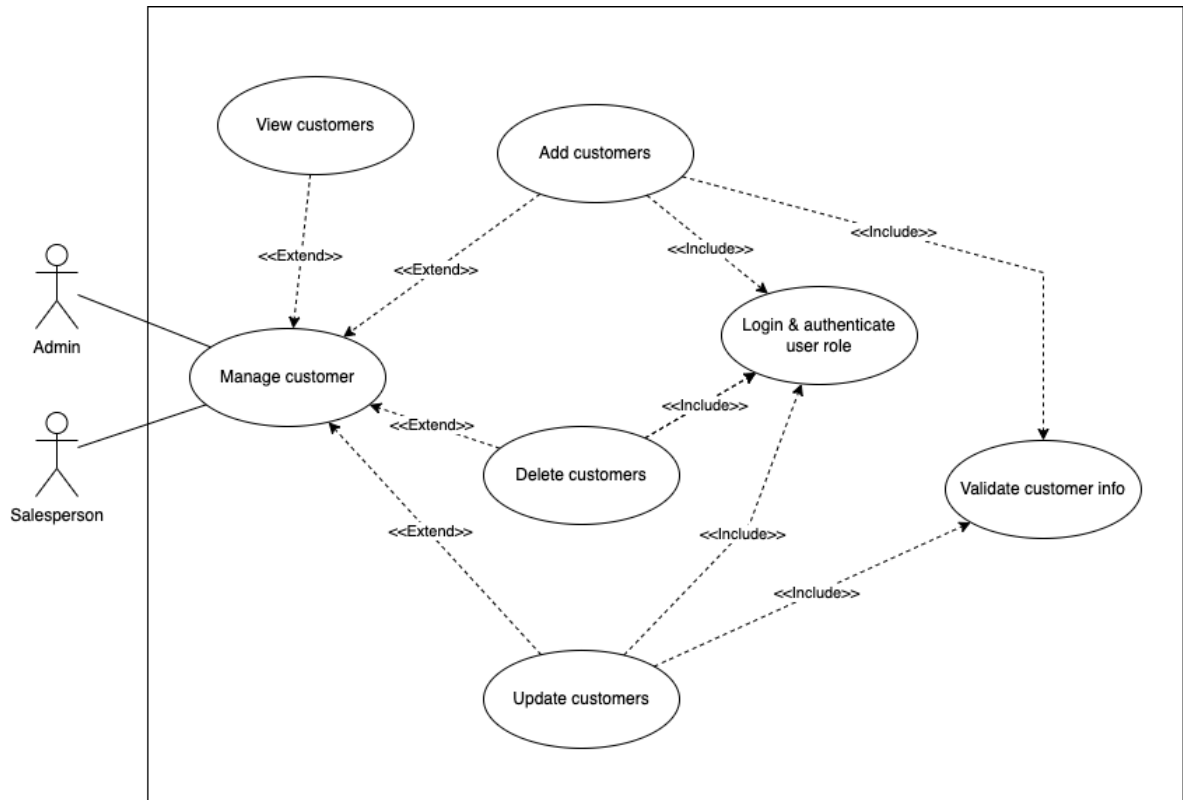
4.2.2 Manage profile use case diagram



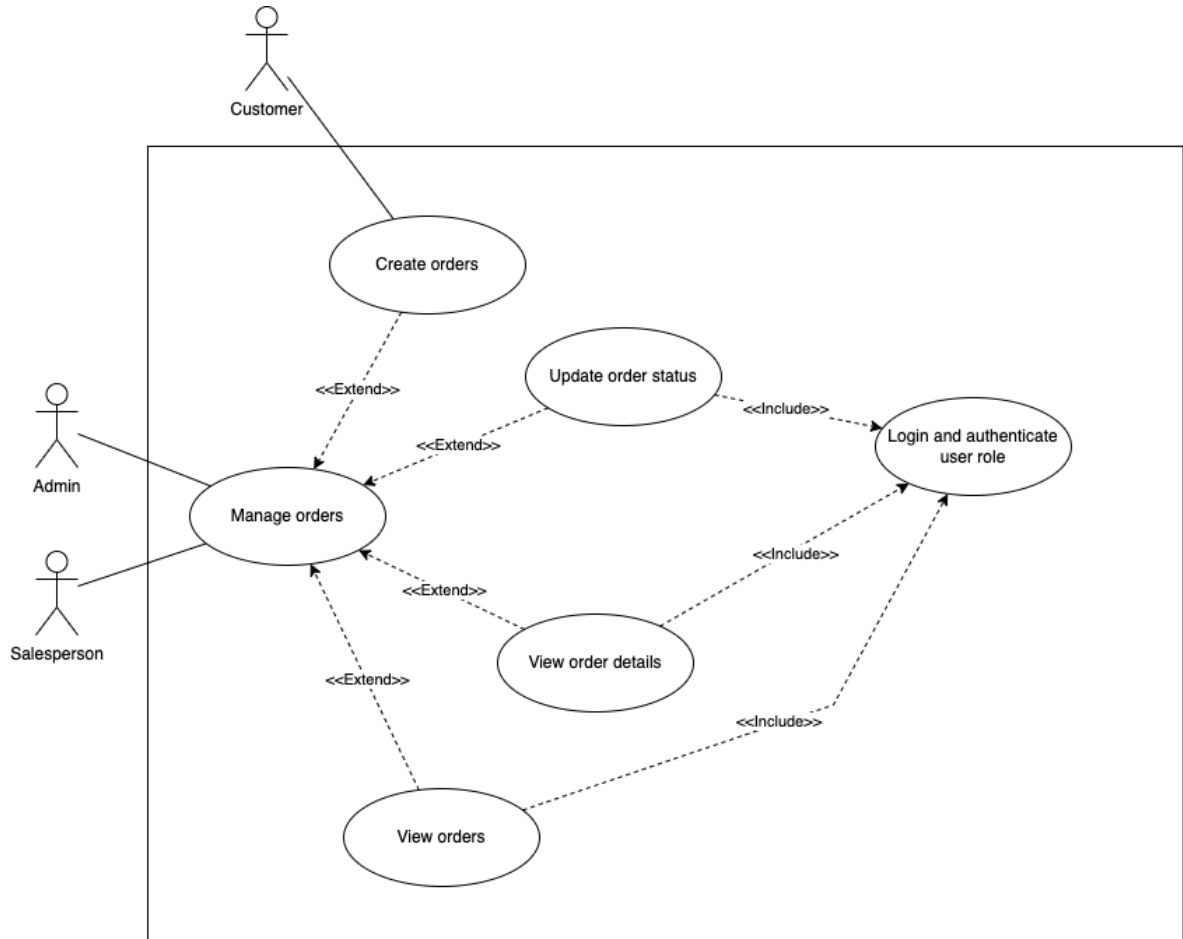
4.2.3 Manage product use case diagram



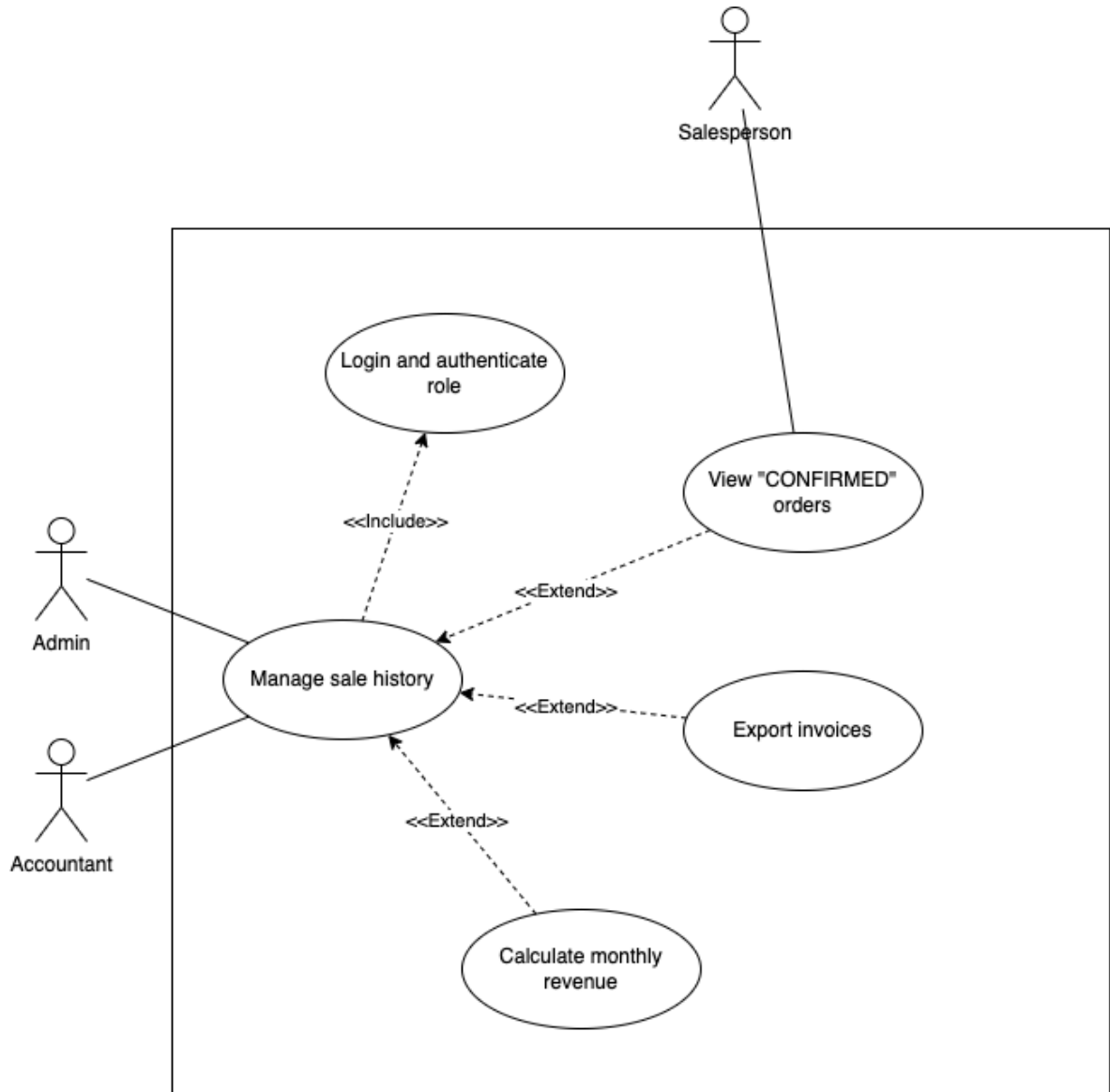
4.2.4 Manage customer use case diagram



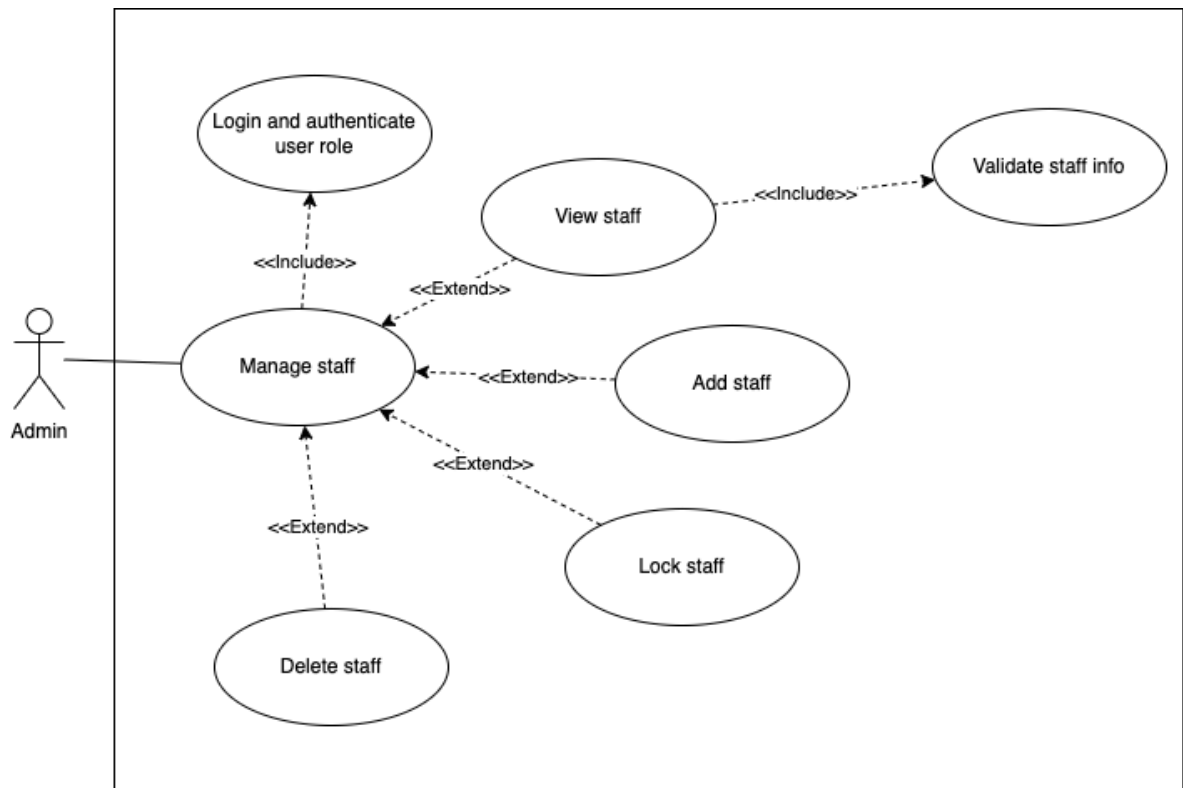
4.2.5 Manage order use case diagram



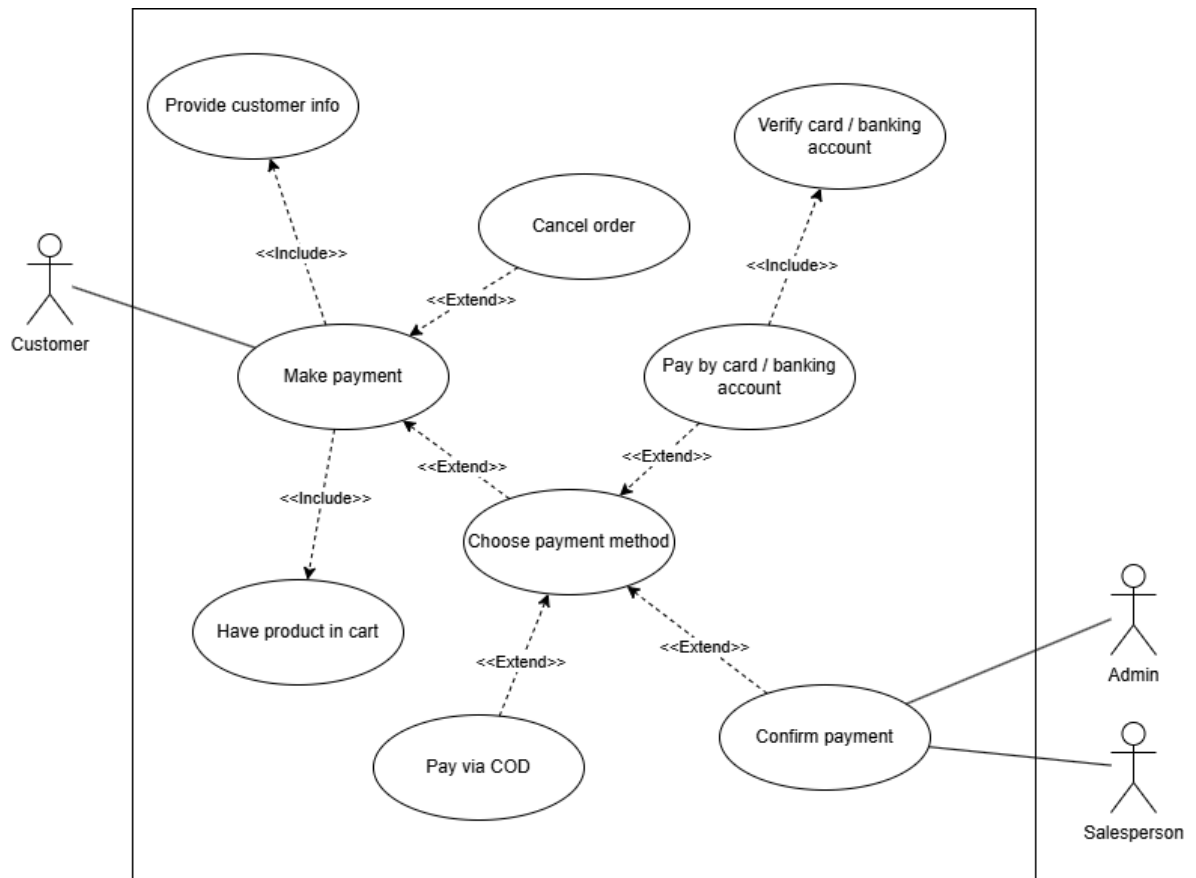
4.2.6 Manage sale history use case diagram



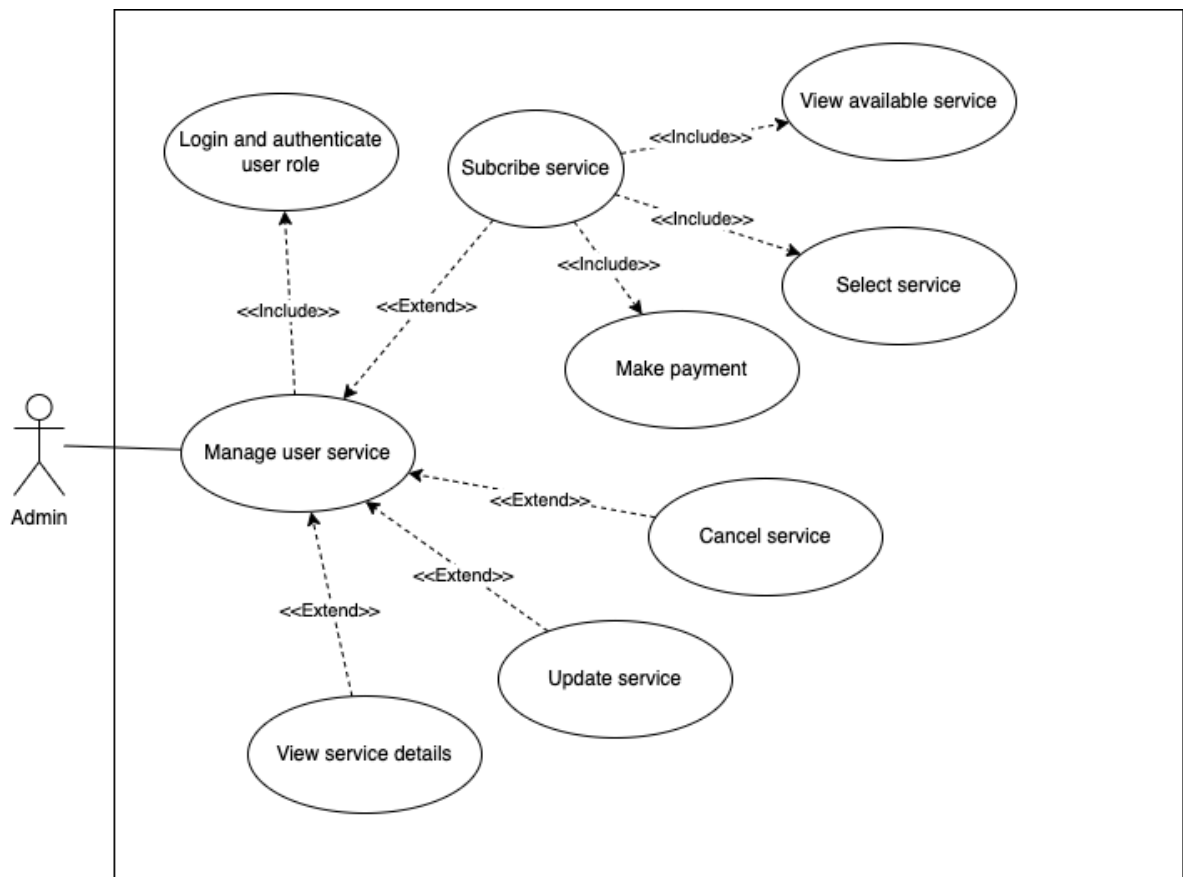
4.2.7 Manage staff use case diagram



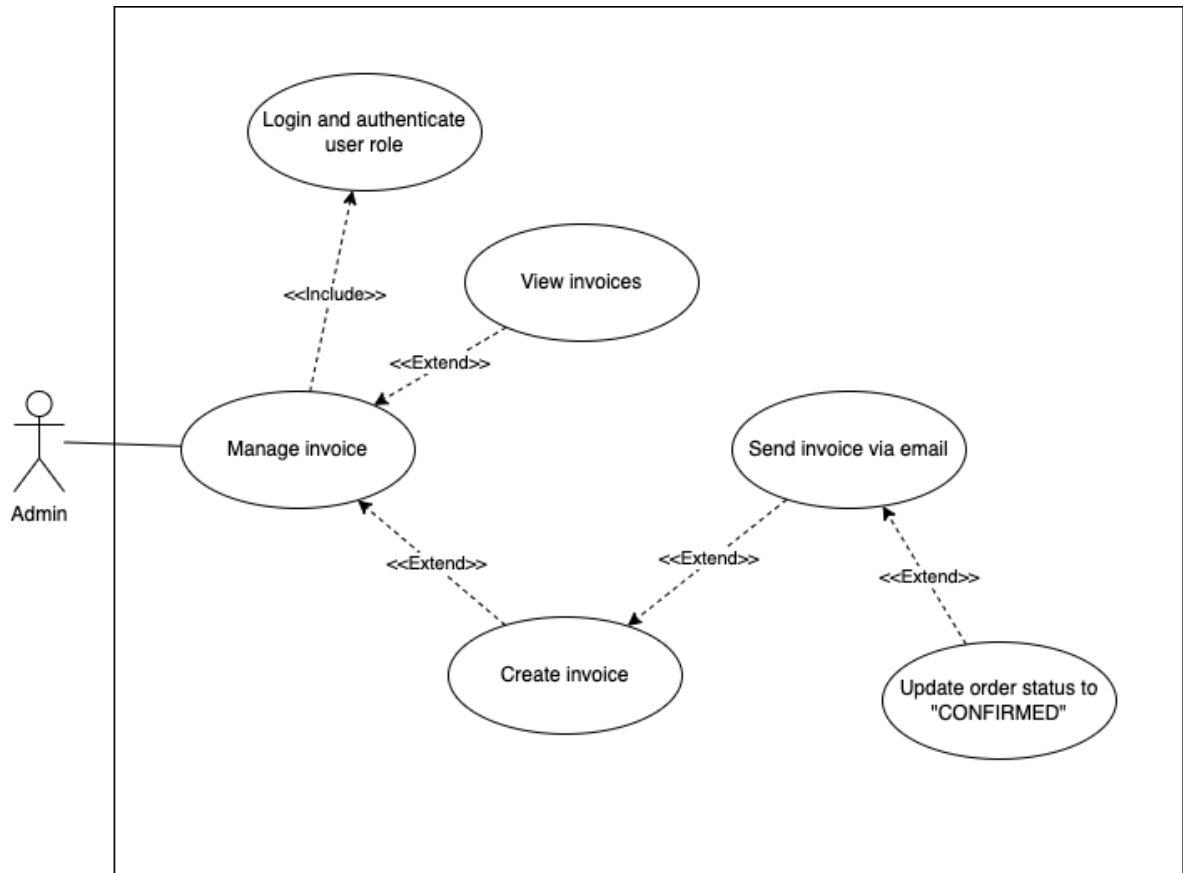
4.2.8 Make payment use case diagram



4.2.9 Manage user service use case diagram



4.2.10 Manage invoice use case diagram



4.3 Use Case Specifications

4.3.1 *Add product*

Field	Details
Use Case Name	Add Product
Scenario	Admin adds a new product to the catalog.
Triggering Event	Admin clicks "Add Product" button.
Brief Description	Admin fills a form to add a new product, and the system saves it.
Actors	Admin
Stakeholders	Admin, Sales Team, Customers
Preconditions	Admin is authenticated and has permission to manage products.
Postconditions	A new product is added to the product catalog.

Flow of Events

Actor	System
1. Admin selects "Add Product" option.	1.1 System displays the product entry form.
2. Admin enters product details.	
3. Admin submits the product form.	3.1 System validates the input data.
	3.2 System checks for duplicate product (by name/SKU).
	3.3 System saves the new product to the catalog.
	3.4 System confirms product creation to Admin.

Exception Conditions

	Condition
1	If required fields are missing, system highlights them and prompts for completion.
2	If product already exists (duplicate name/SKU), system blocks creation and notifies admin.
3	If invalid format is entered (e.g. negative price), system shows validation error.
4	If admin has insufficient permission, system blocks access to the feature.

4.3.2 *Update Product*

Field	Details
Use Case Name	Update Product
Scenario	Admin updates details of an existing product.
Triggering Event	Admin selects a product from the catalog and clicks "Edit".
Brief Description	Allows admin to change name, price, description, image, or availability.
Actors	Admin
Stakeholders	Admin, Sales Team, Customers
Preconditions	Product exists and admin is authenticated with edit permission.
Postconditions	The product is updated with the modified details.

Flow of Events

Actor	System
1. Admin selects a product to edit.	1.1 System loads current product details into the form.
2. Admin modifies product details.	
3. Admin submits the updated form.	3.1 System validates the input fields.
	3.2 System checks for conflicts or invalid entries.
	3.3 System updates product data in the database.
	3.4 System confirms the update to admin.

Exception Conditions

	Condition
1	If product does not exist, system displays error message.
2	If input is invalid (e.g. price < 0), system shows validation error.
3	If product is under active order, system may restrict certain fields from edit.
4	If admin lacks permission, system blocks update and notifies the user.

4.3.3 Delete Product

Field	Details
Use Case Name	Delete Product
Scenario	Admin deletes a product from the system.
Triggering Event	Admin clicks the "Delete" button on a selected product.
Brief Description	Removes the selected product from the catalog.
Actors	Admin
Stakeholders	Admin, Sales
Preconditions	Admin has delete permission and product exists.
Postconditions	Product is removed from the product list.

Flow of Events

Actor	System
1. Admin selects a product to delete.	1.1 System prompts for deletion confirmation.
2. Admin confirms deletion.	2.1 System verifies that product is not linked to any open order.
	2.2 System deletes product from the catalog.
	2.3 System displays deletion success message.

Exception Conditions

	Condition
1	If product is associated with active orders, deletion is blocked.
2	If product does not exist, system returns an error.
3	If admin cancels at confirmation, product is not deleted.
4	If admin lacks permission, system blocks deletion action.

4.3.4 Add Staff

Use Case Name	Add Staff
Scenario	Admin adds a new staff member to the system.
Triggering Event	Admin clicks "Add Staff" on the user management dashboard.
Brief Description	Admin enters staff details and assigns a role. System validates and creates the user.
Actors	Admin
Stakeholders	HR Department: to manage employee records IT Department: to handle access rights
Preconditions	Admin is authenticated and has the "manage users" permission.
Postconditions	New staff member is added and granted system access based on assigned role.

Flow of Events

Actor	System
1. Admin selects "Add Staff" from the user management dashboard.	1.1 System displays the staff creation form.
2. Admin enters staff details (name, email, phone, role, etc.).	
3. Admin clicks the "Submit" button.	3.1 System validates the entered data.
	3.2 System checks for duplicate email/username.
	3.3 System creates a new staff account.
	3.4 System sends confirmation or welcome email (optional).
	3.5 System displays a success message to the admin.

Exception Conditions

	Condition
1.1	If required fields are missing, the system highlights them and prompts admin to complete input.
1.2	If the entered email or username already exists, the system shows a duplication warning.
1.3	If input format is invalid (e.g., invalid email or phone), the system displays error message.
1.4	If admin lacks permission, access is denied and the use case is aborted.

4.3.5 *Update Staff*

Use Case Name	Update Staff
Scenario	Admin updates the information of an existing staff member.
Triggering Event	Admin selects a staff profile and clicks "Edit".
Brief Description	Allows admin to update staff name, role, contact, or status.
Actors	Admin
Stakeholders	HR, Admin
Preconditions	Admin is authenticated and the staff account exists.
Postconditions	Staff data is updated and changes reflected in the system.

Flow of Events

Actor	System
1. Admin opens staff management panel.	1.1 System loads and displays staff list.
2. Admin selects a staff and clicks "Edit".	2.1 System displays current information in editable form.
3. Admin modifies relevant details (e.g., role, contact info).	
4. Admin clicks "Save" or "Update".	4.1 System validates new input.
	4.2 System checks business rules (e.g., cannot downgrade super admin).
	4.3 System updates staff record in database.
	4.4 System displays success message.

Exception Conditions

	Condition
1.1	Staff not found (deleted or invalid ID).
1.2	Missing or invalid input (e.g., invalid phone format).
1.3	Attempt to assign invalid role or permission.
1.4	Admin lacks sufficient rights to update target staff (e.g., same/higher privilege level).

4.3.6 Delete Staff

Use Case Name	Delete Staff
Scenario	Admin deletes a staff account from the system.
Triggering Event	Admin selects a staff and clicks the "Delete" button.
Brief Description	Permanently removes a staff account unless it's protected or critical.
Actors	Admin
Stakeholders	HR, Admin
Preconditions	Staff exists, admin is authorized to delete.
Postconditions	Staff account is removed from user database.

Flow of Events

Actor	System
1. Admin opens the staff list.	1.1 System displays all active staff accounts.
2. Admin selects a staff and clicks "Delete".	2.1 System prompts for confirmation.
3. Admin confirms deletion.	3.1 System checks if staff is deletable (not system-critical).
	3.2 System removes staff from database.
	3.3 System displays confirmation of successful deletion.

Exception Conditions

	Condition
1.1	Staff not found or already deleted.
1.2	Staff is system-critical (e.g., sole admin) and cannot be deleted.
1.3	Admin lacks necessary permission to delete staff.
1.4	Admin cancels the confirmation dialog.

4.3.7 *Add Customer*

Use Case Name	Add Customer
Scenario	Admin or Salesperson adds a new customer to the system.
Triggering Event	User clicks "Add Customer" in the customer management section.
Brief Description	Captures customer profile details and creates a new customer record.
Actors	Admin, Salesperson
Stakeholders	Sales Department, Customer Support, Marketing
Preconditions	User is authenticated with permission to manage customers.
Postconditions	New customer profile is saved and available in the system.

Flow of Events

Actor	System
1. User clicks "Add Customer".	1.1 System displays the customer registration form.
2. User fills in customer information.	
3. User submits the form.	3.1 System validates the data.
	3.2 System checks for duplicates by email/phone.
	3.3 System creates a new customer record in the database.
	3.4 System displays confirmation message.

Exception Conditions

	Condition
1.1	Required fields are missing; system highlights the missing fields.
1.2	Duplicate customer found (by email or phone); system shows a duplication warning.
1.3	Input format invalid (e.g., invalid email or phone).
1.4	User lacks permission to add customer.

4.3.8 *Update customer info*

Use Case Name	Update Customer Info
Scenario	Admin or Salesperson edits an existing customer's information.
Triggering Event	User selects a customer profile and clicks "Edit".
Brief Description	Updates customer name, contact, status, or preferences.
Actors	Admin, Salesperson
Stakeholders	Customer Support, Sales, Marketing
Preconditions	Customer exists; user is authorized to modify customer records.
Postconditions	Customer information is updated in the system.

Flow of Events

Actor	System
1. User selects a customer to edit.	1.1 System loads the customer's current data into editable fields.
2. User updates relevant information.	
3. User submits the updated form.	3.1 System validates the input fields.
	3.2 System updates the customer record in the database.
	3.3 System confirms the update to the user.

Exception Conditions

	Condition
1.1	Customer not found or deleted.
1.2	Invalid input format (e.g., phone or email).
1.3	Attempt to update restricted fields (e.g., internal CRM ID).
1.4	Insufficient permissions to update data.

4.3.9 *Cancel Order*

Use Case Name	Cancel Order
Scenario	Customer cancels an active order.
Triggering Event	User clicks "Cancel Order" on the order details screen.
Brief Description	Cancels the order before it is processed or shipped.
Actors	Customer
Stakeholders	Customer, Sales, Accounting
Preconditions	Order is in cancellable state (e.g., not shipped).
Postconditions	Order status is set to "Cancelled" and no further processing occurs.

Flow of Events

Actor	System
1. Customer views order list.	1.1 System displays active orders with status and actions.
2. Customer clicks "Cancel".	2.1 System prompts for confirmation.
3. Customer confirms.	3.1 System verifies order status is cancellable.
	3.2 System updates order status to "Cancelled".
	3.3 System logs the action and sends confirmation to customer.

Exception Conditions

	Condition
1.1	Order is already shipped or delivered and cannot be cancelled.
1.2	System error while updating order status.
1.3	Cancellation window expired (e.g., after 24h).
1.4	Unauthorized user attempts cancellation.

4.3.10 Return Order

Use Case Name	Return Order
Scenario	Customer initiates a return for a completed order.
Triggering Event	Customer clicks "Return" on the order history page.
Brief Description	Initiates the return process and creates a return transaction.
Actors	Customer
Stakeholders	Customer, Sales, Logistics, Accounting
Preconditions	Order is marked as delivered and within the return policy period.
Postconditions	Return request is logged and flagged for review or approval.

Flow of Events

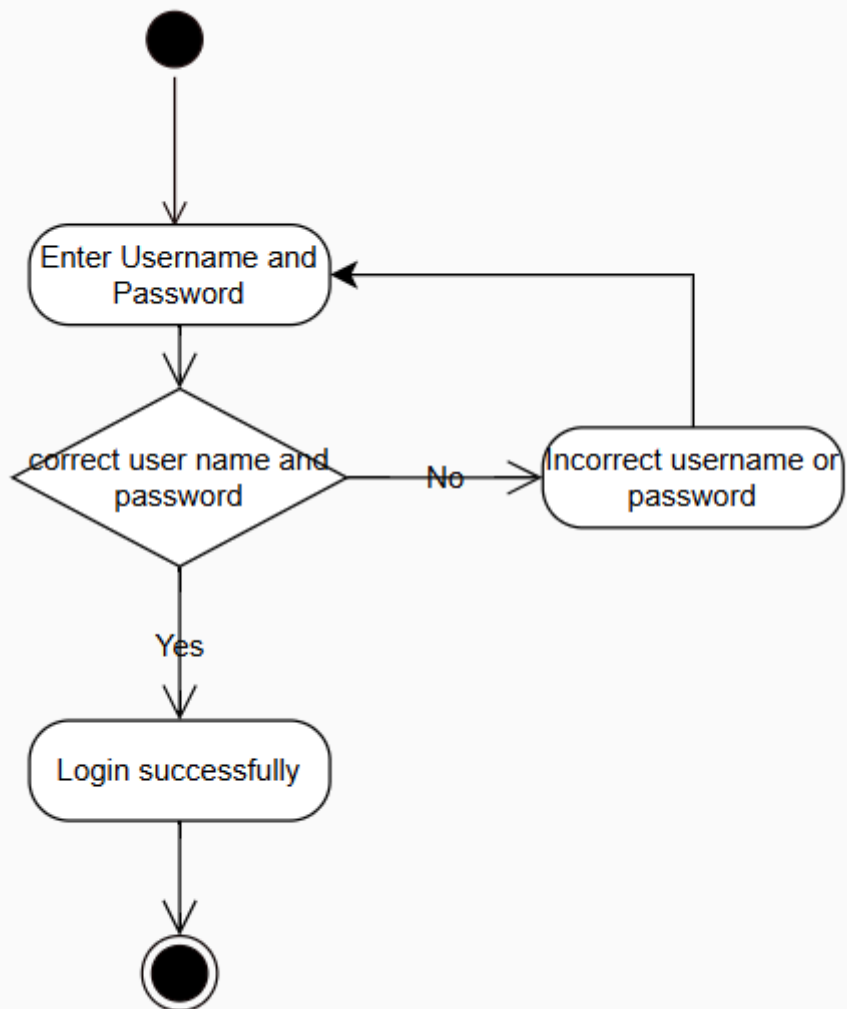
Actor	System
1. Customer opens order history.	1.1 System displays all completed orders.
2. Customer clicks "Return" on an item.	2.1 System prompts for return reason and quantity.
3. Customer submits return request.	3.1 System validates if return is allowed (within policy, timeframe).
	3.2 System creates return request entry.
	3.3 System sends confirmation or return instructions.

Exception Conditions

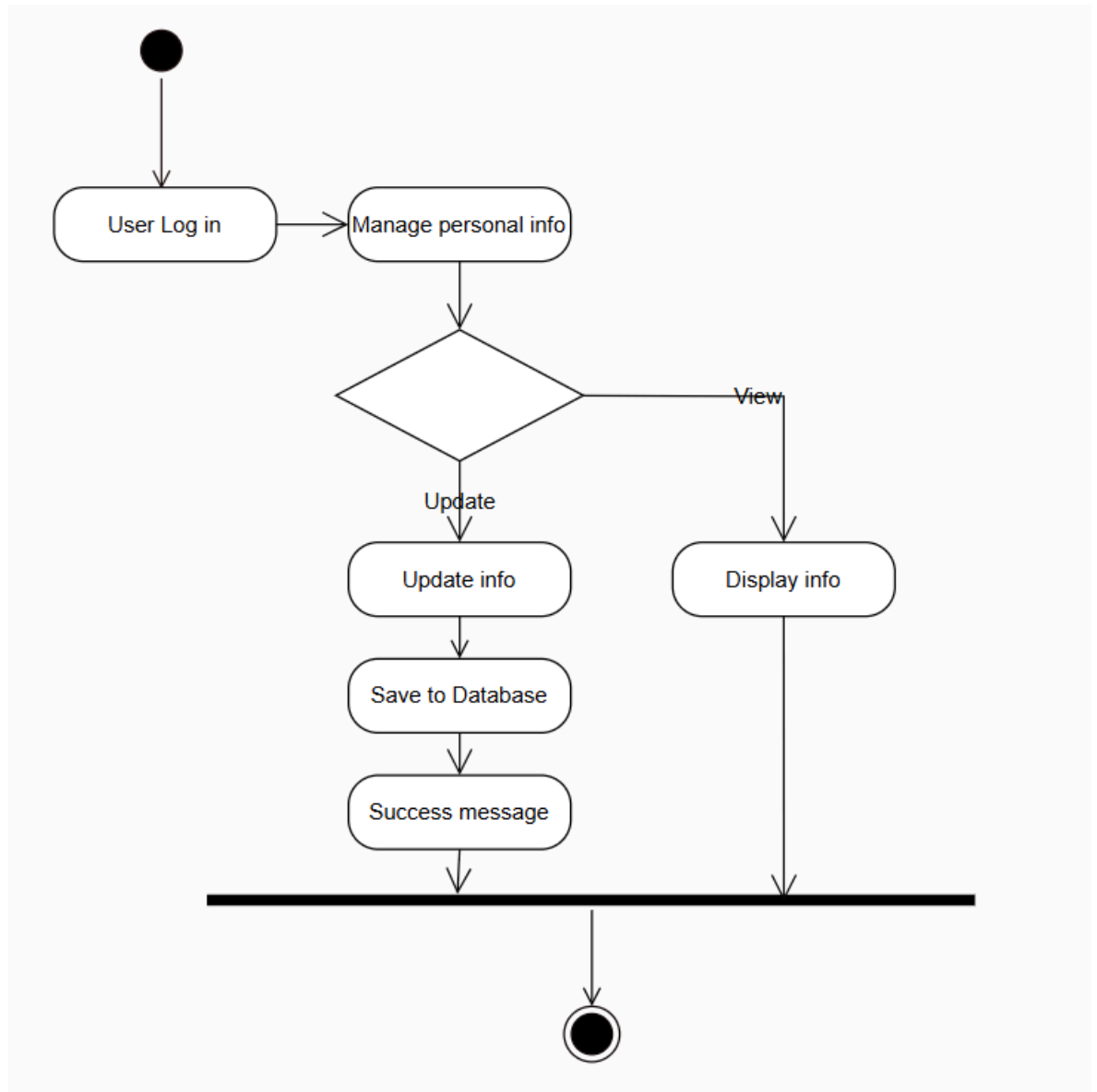
	Condition
1.1	Return period expired.
1.2	Item is marked as non-returnable.
1.3	Customer submits invalid reason/quantity.
1.4	Return system or backend failure during request creation.

CHAPTER 5. ACTIVITY DIAGRAM

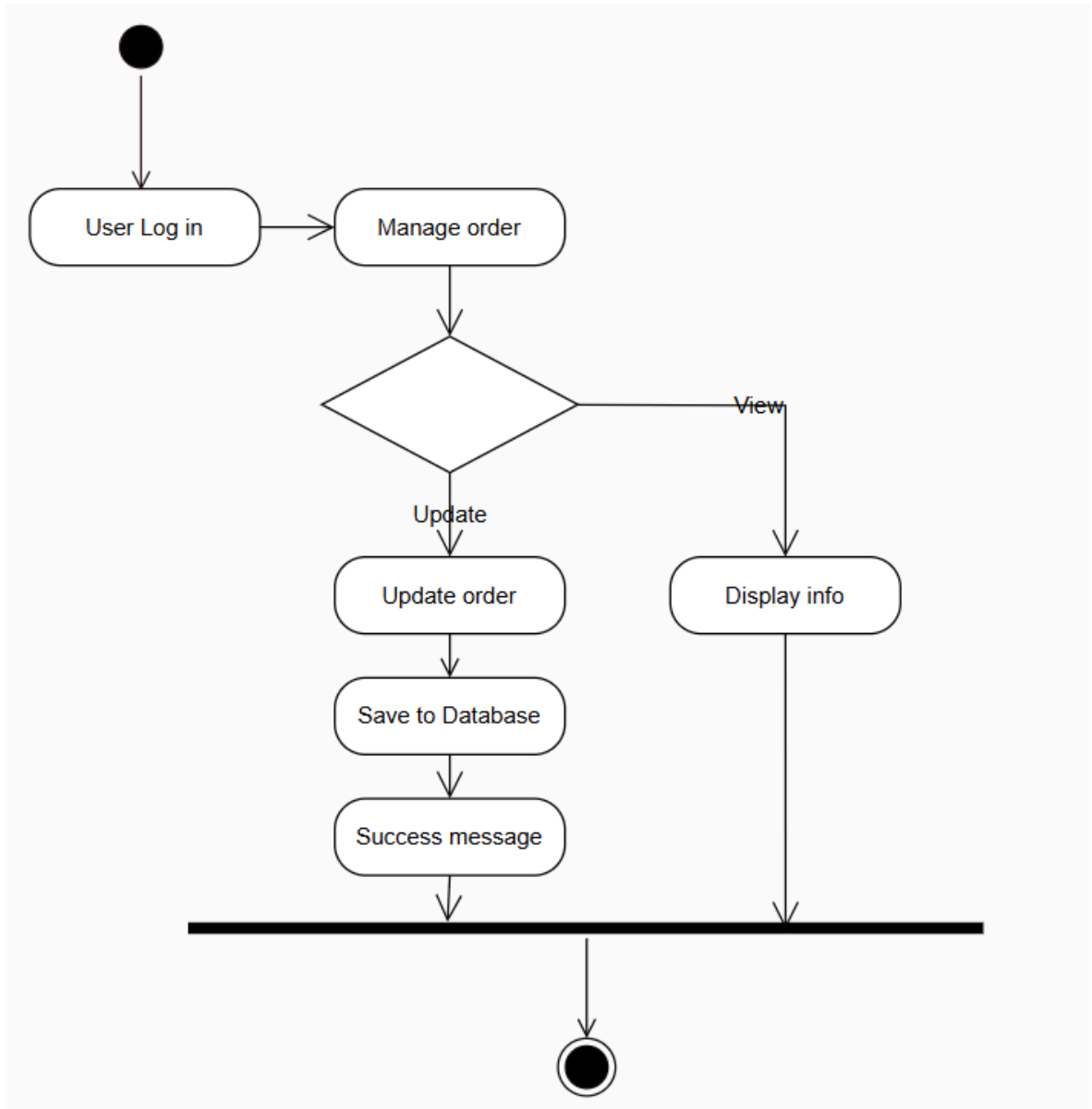
5.1 Login



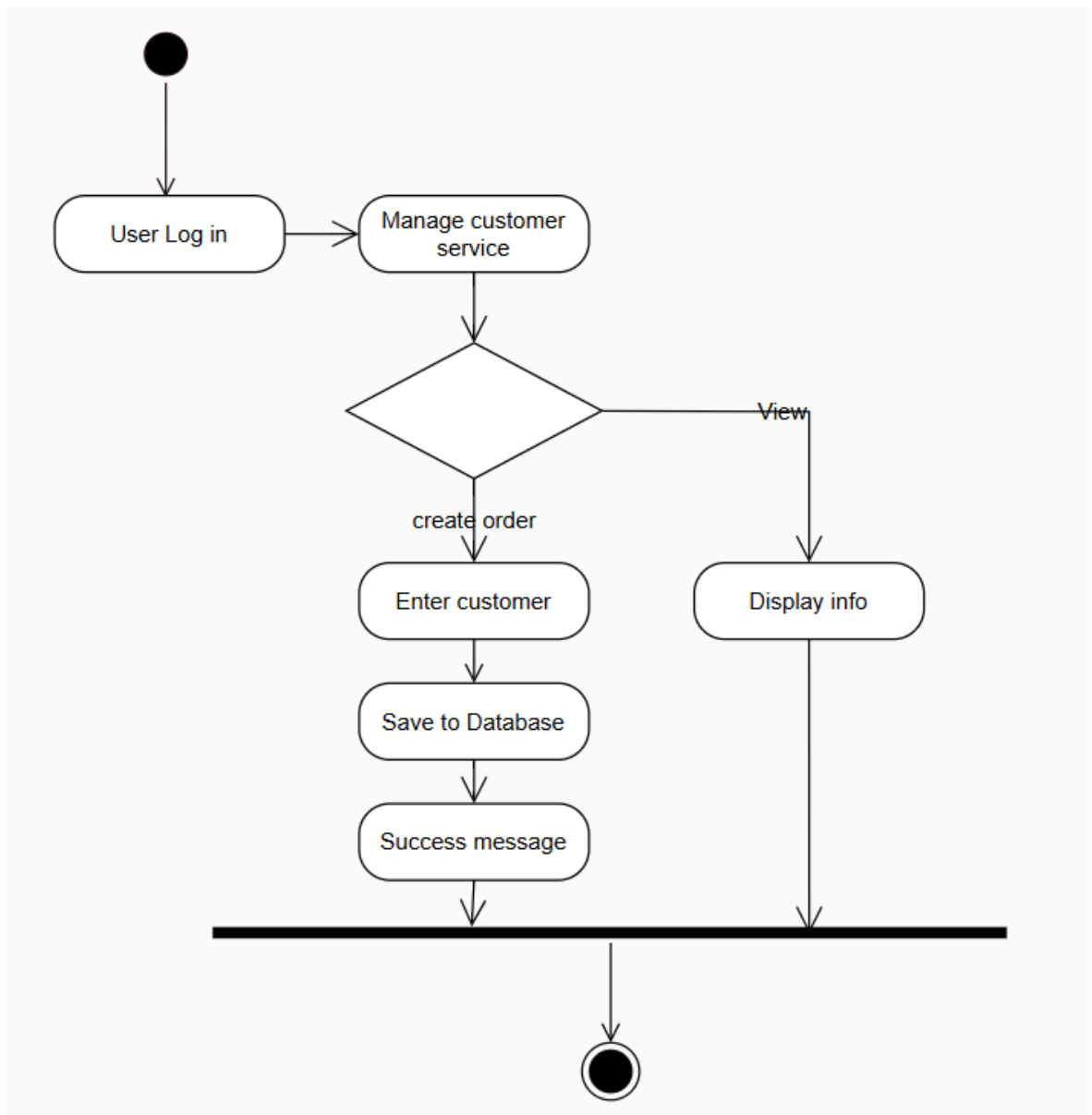
5.2 Manage personal info



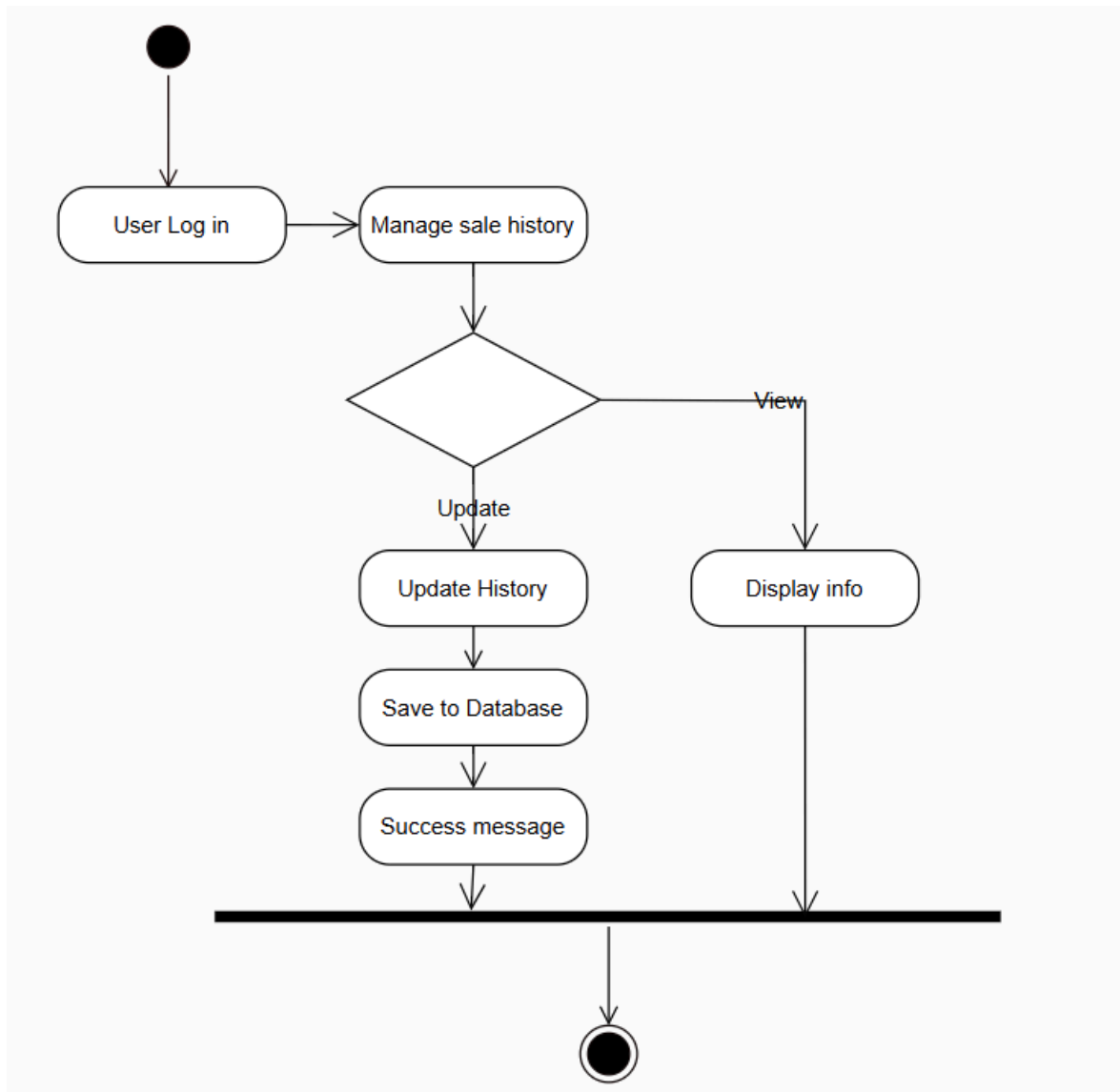
5.3 Manage order



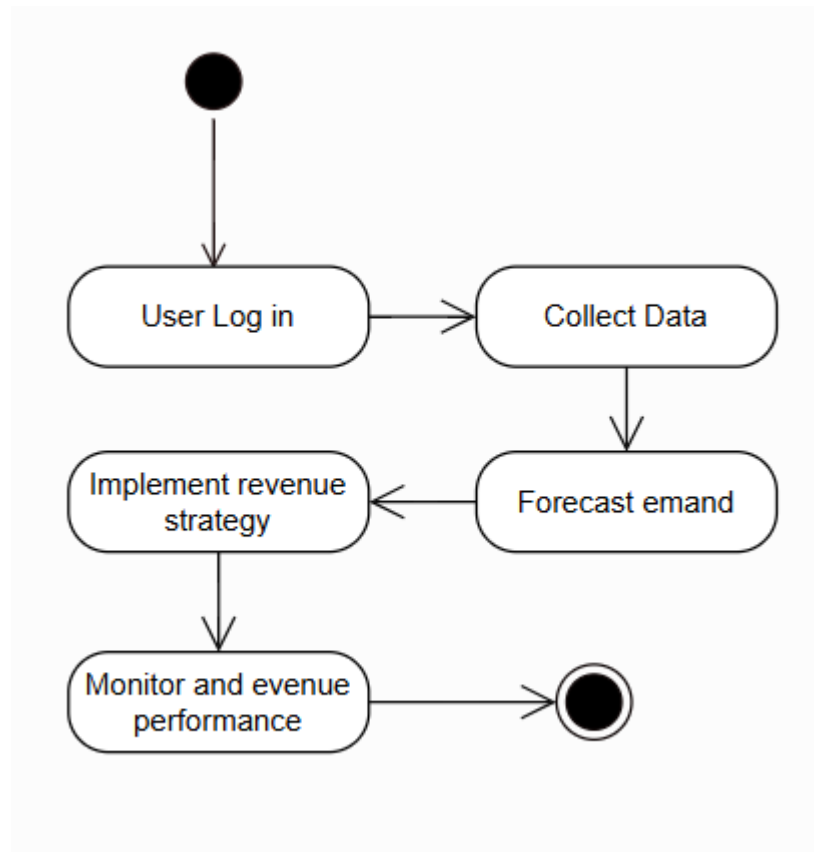
5.4 Manage customer service



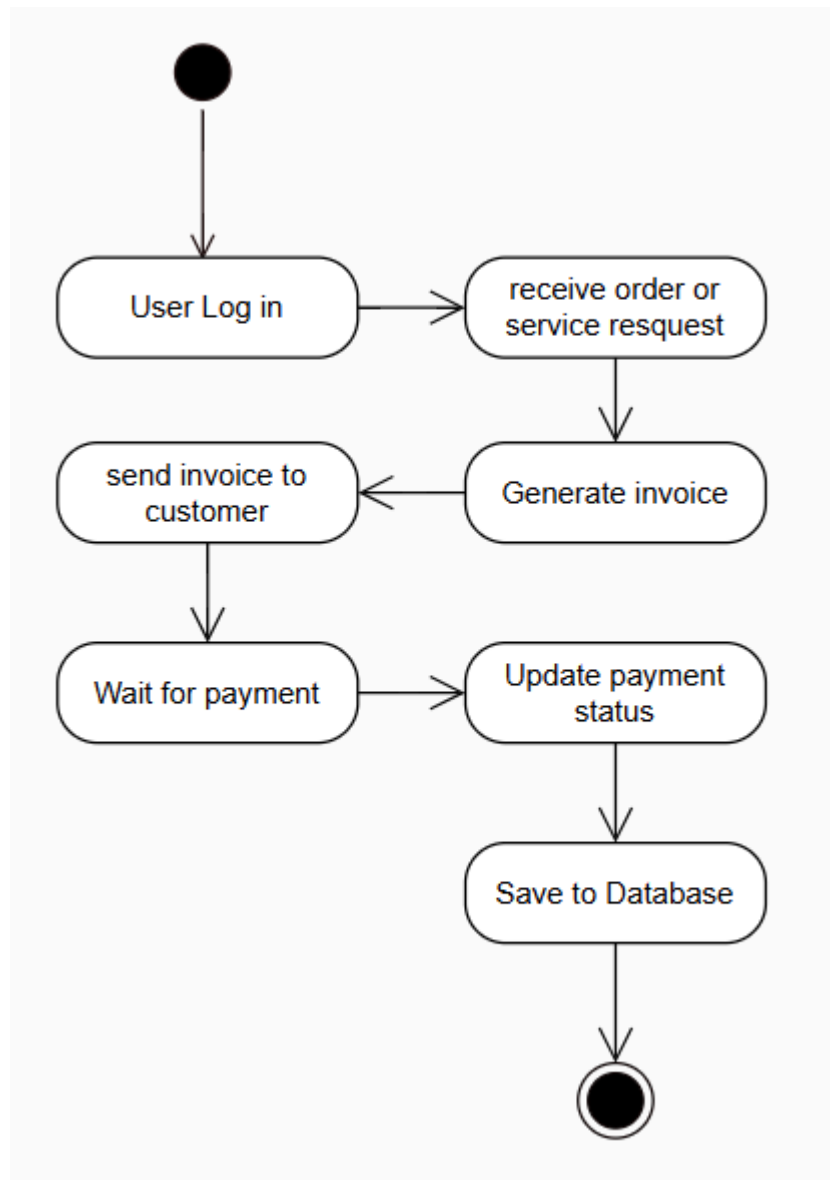
5.5 Manage sale history



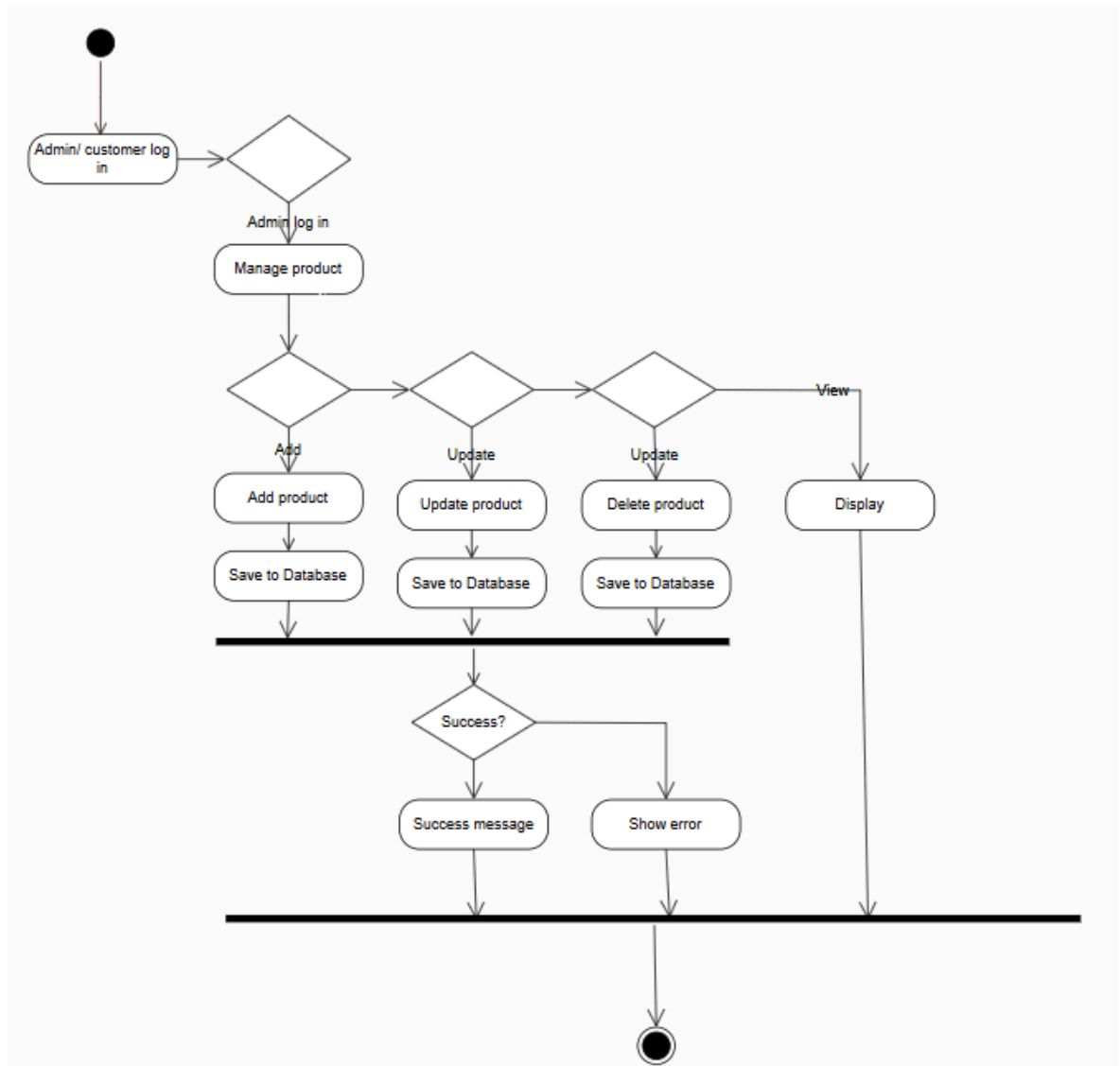
5.6 Manage revenue



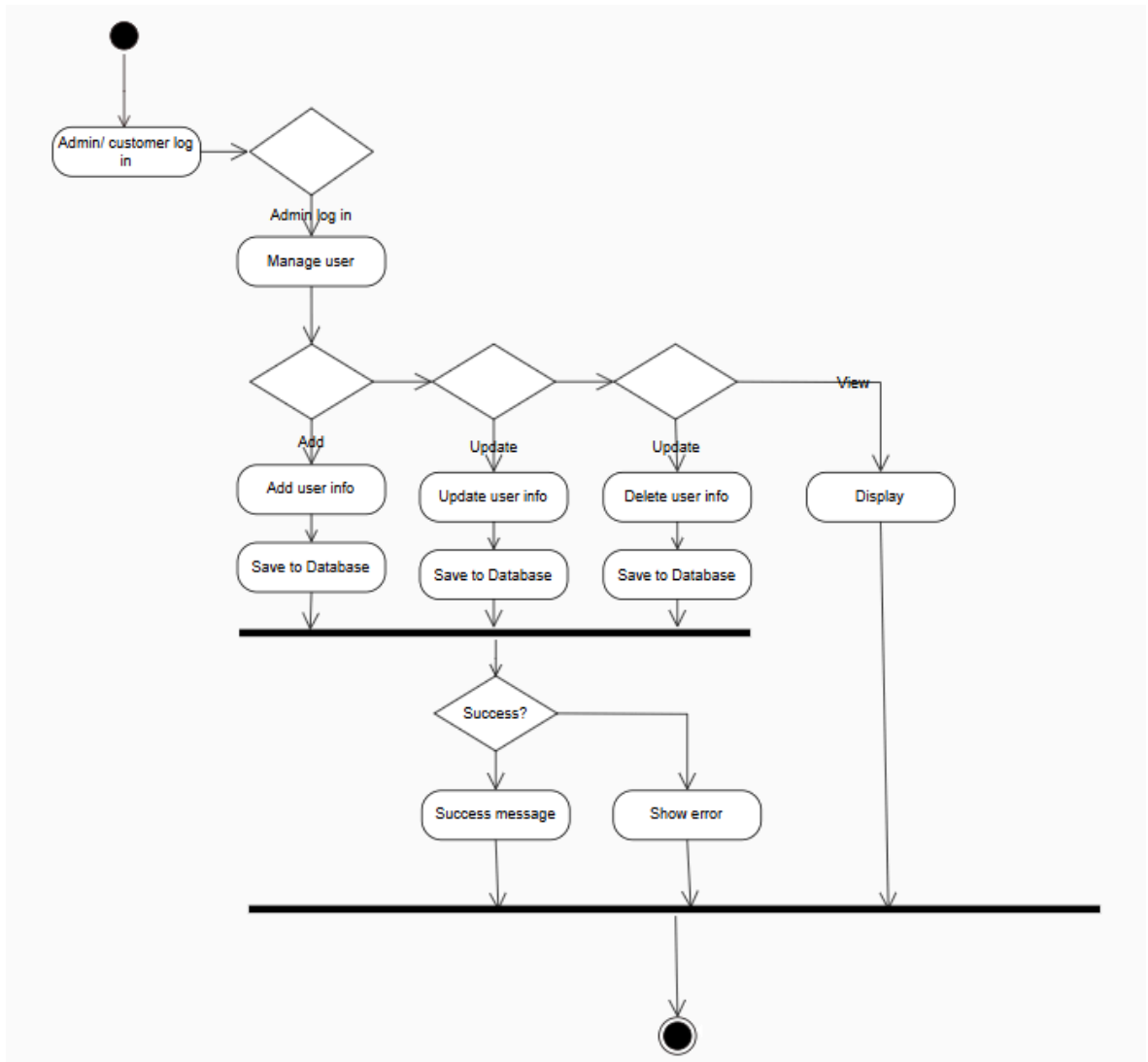
5.7 Manage invoices



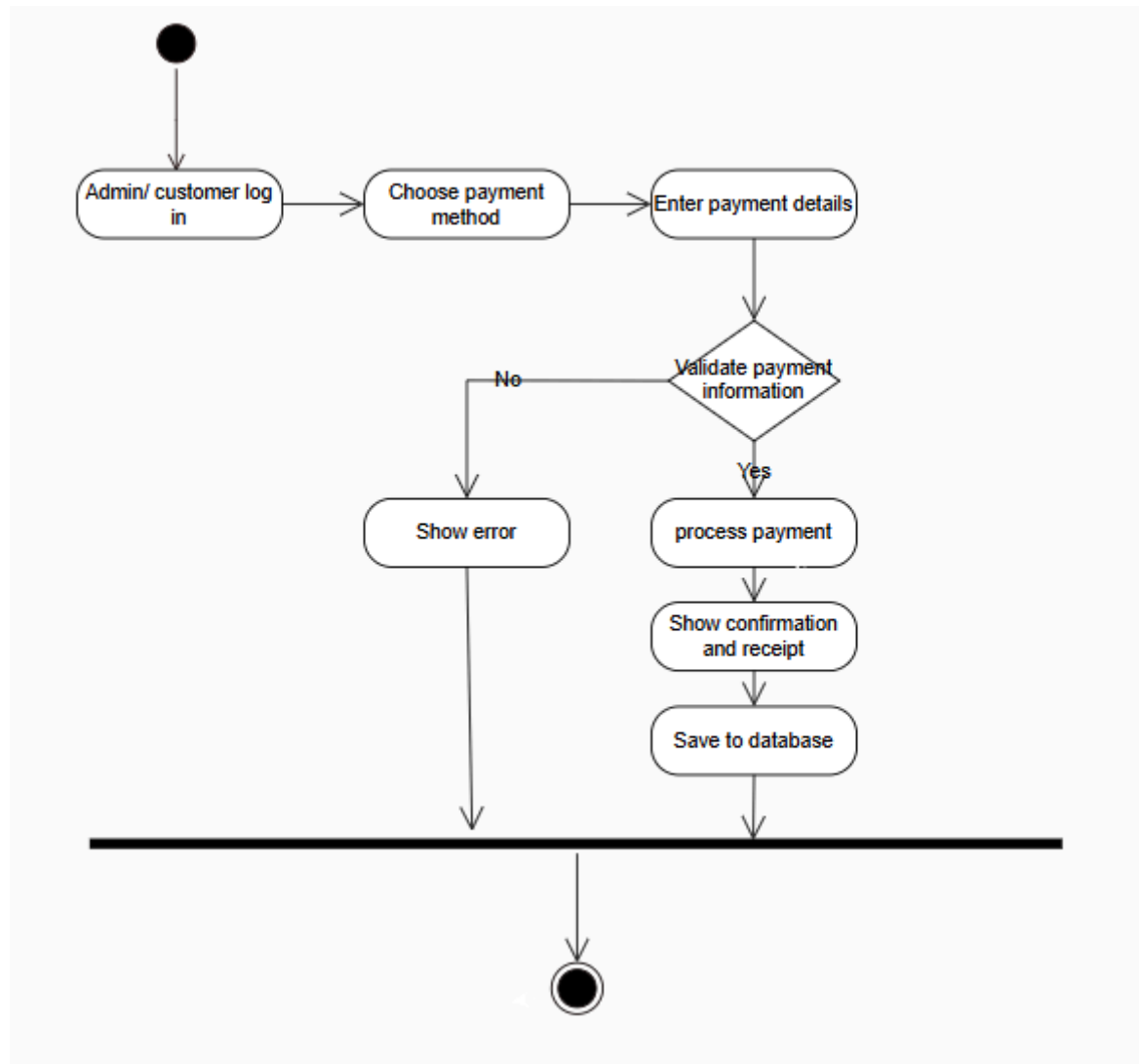
5.8 Manage products



5.9 Manage users

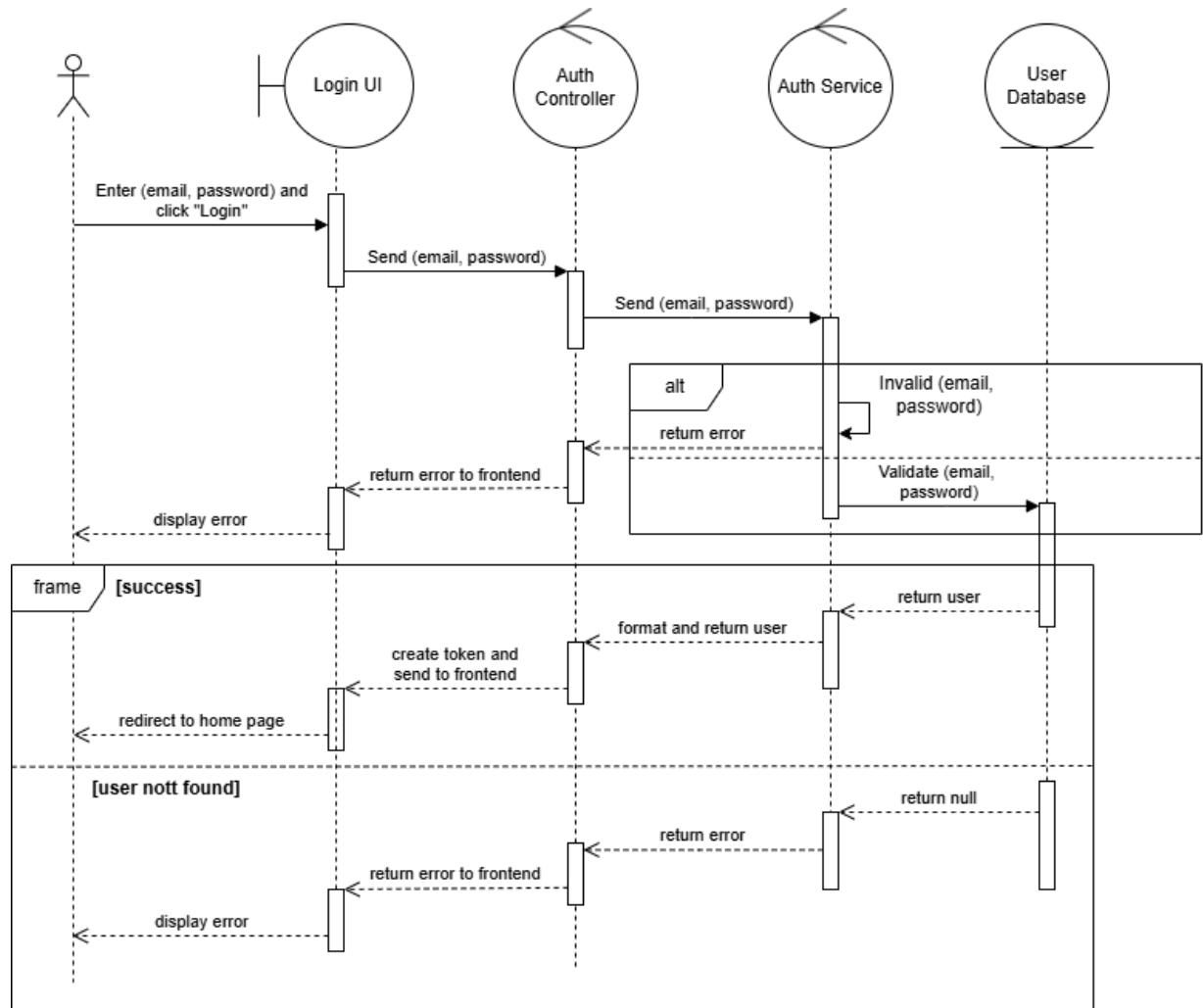


5.10 Manage payment

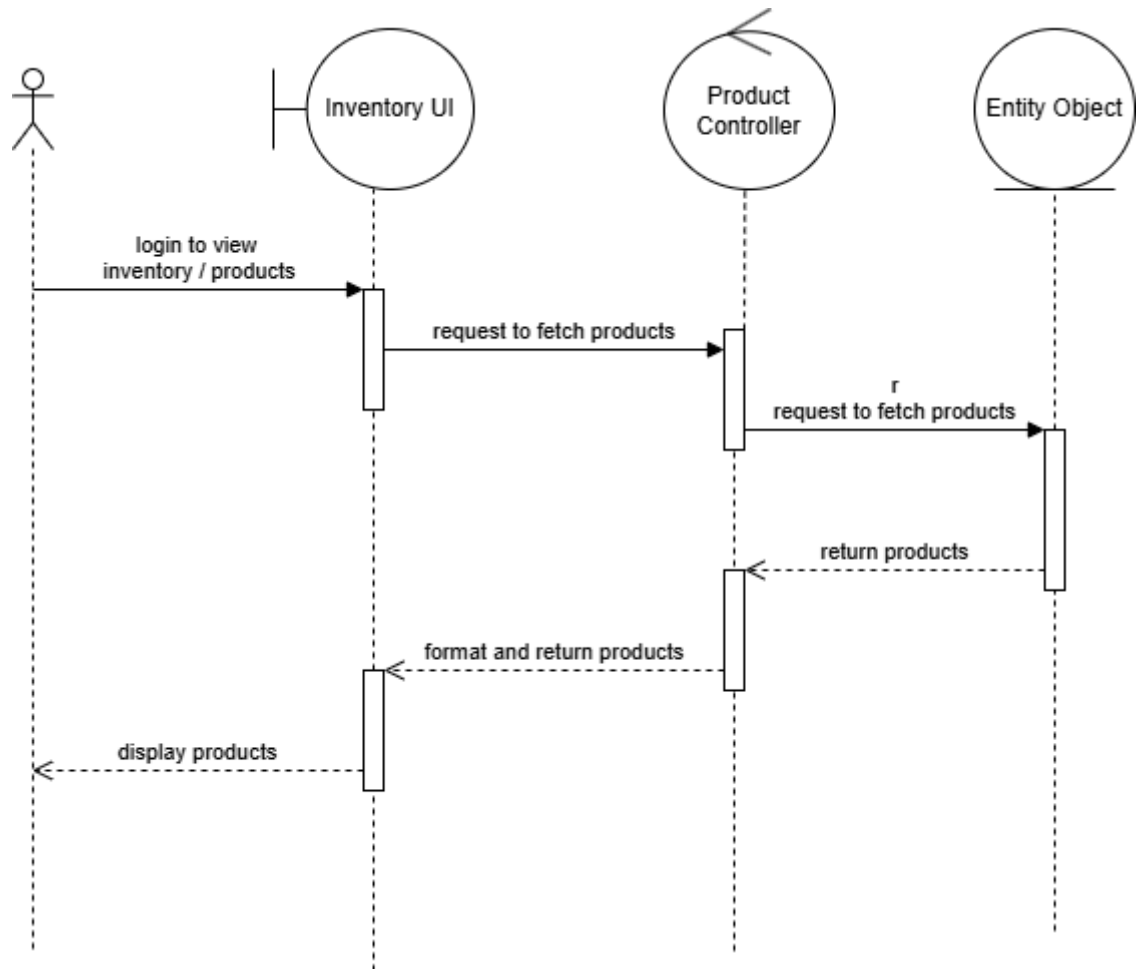


CHAPTER 6. SEQUENCE DIAGRAM

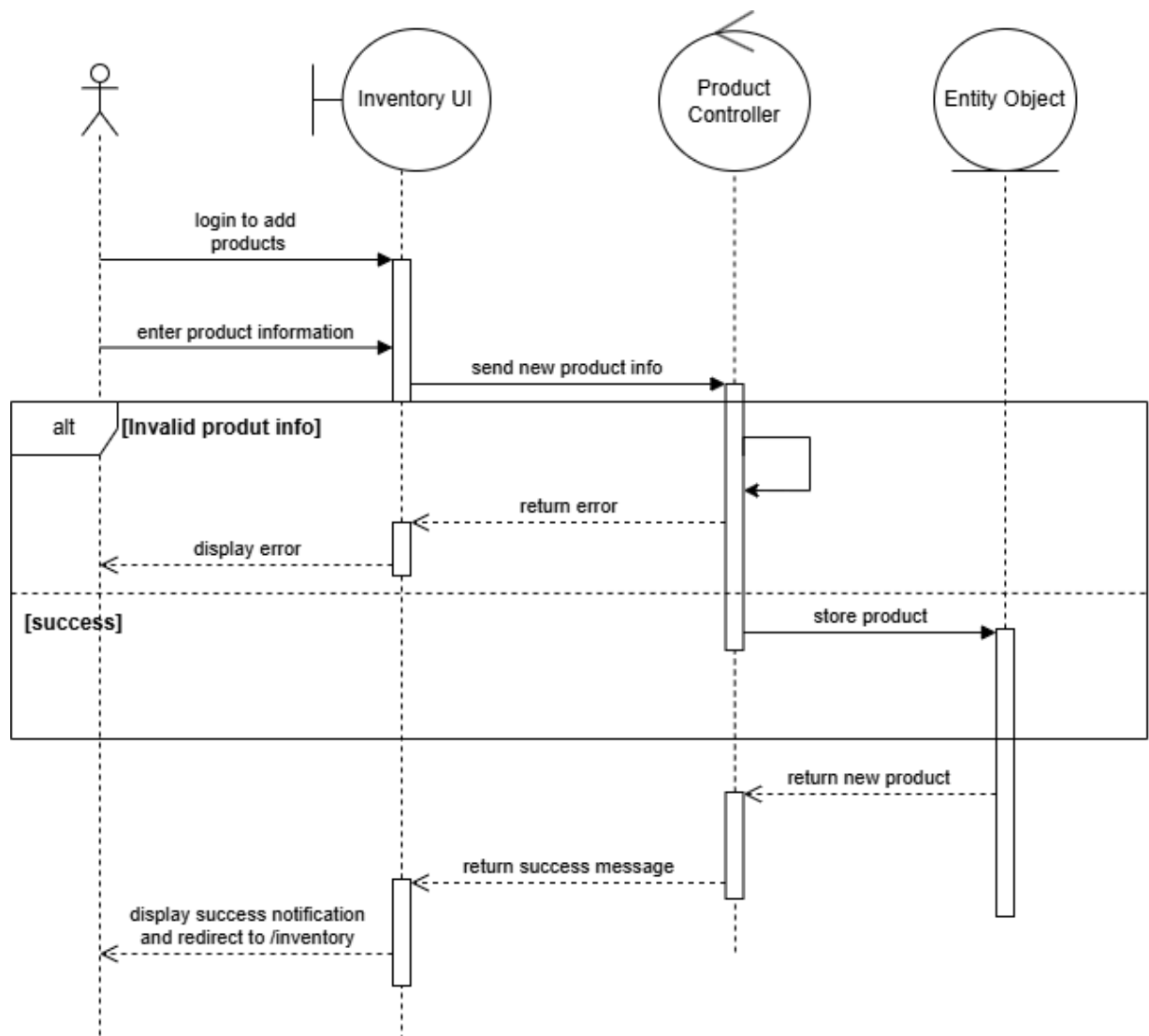
6.1 Login sequence diagram



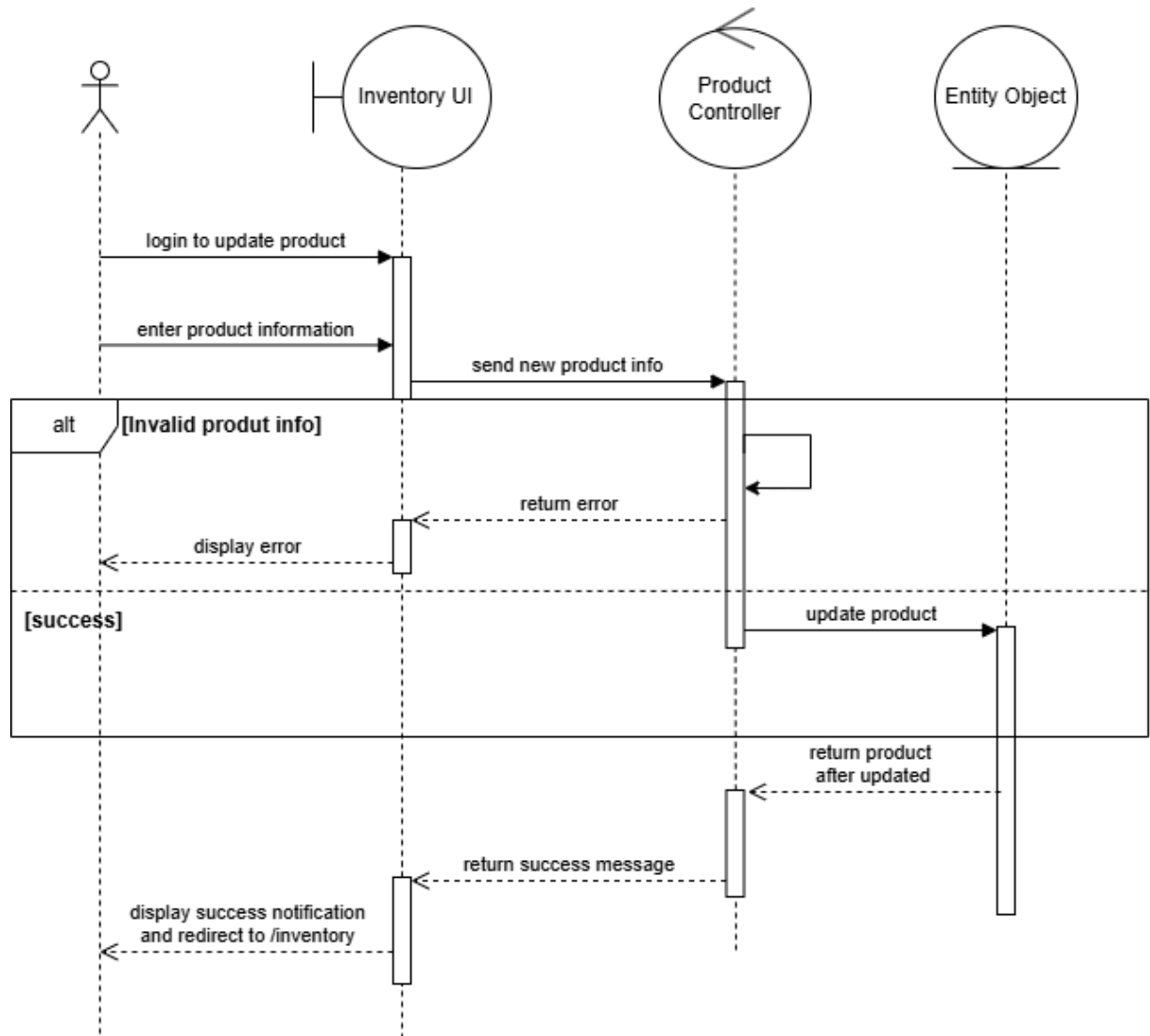
6.2 View inventory sequence diagram



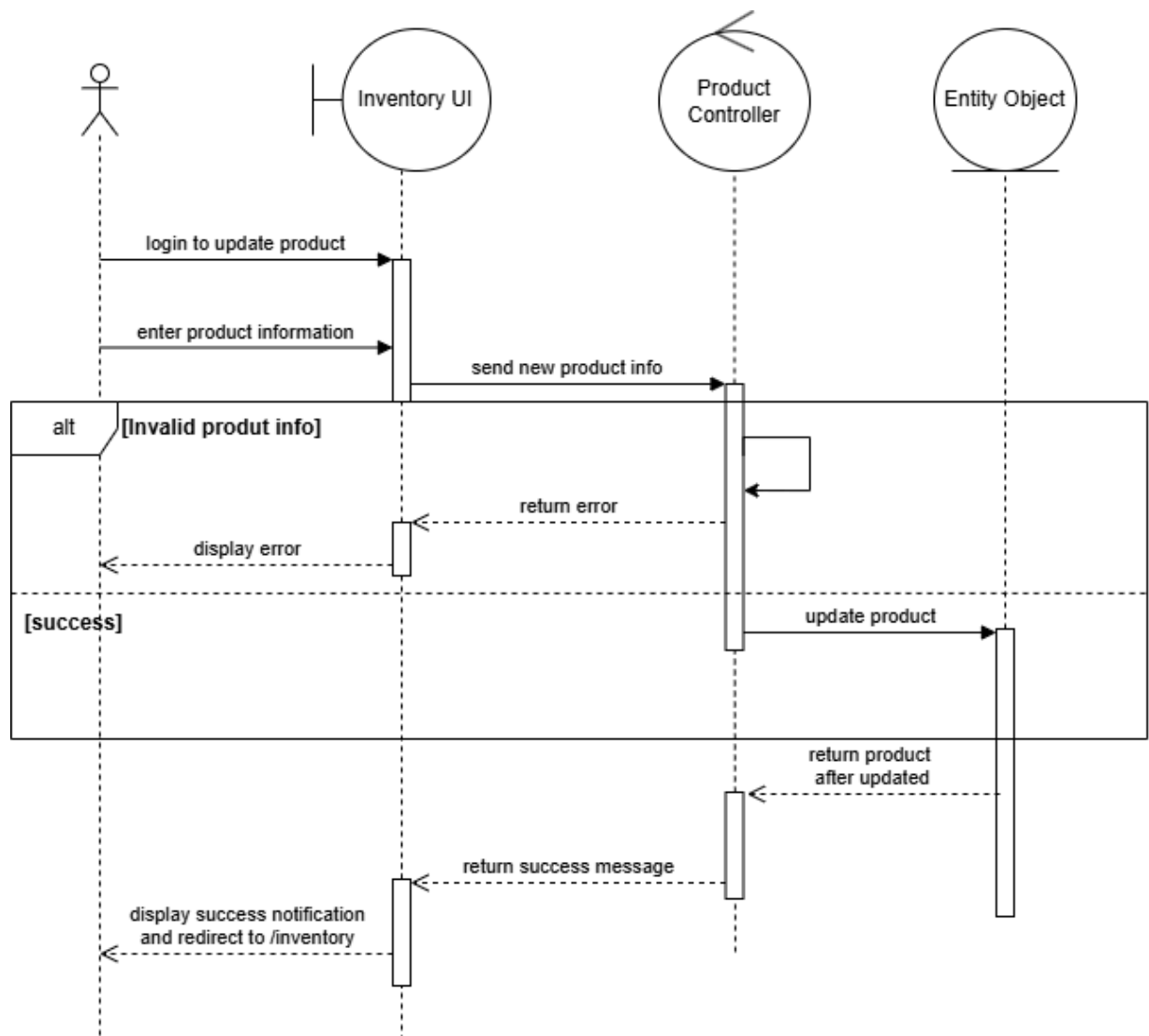
6.3 Add product sequence diagram



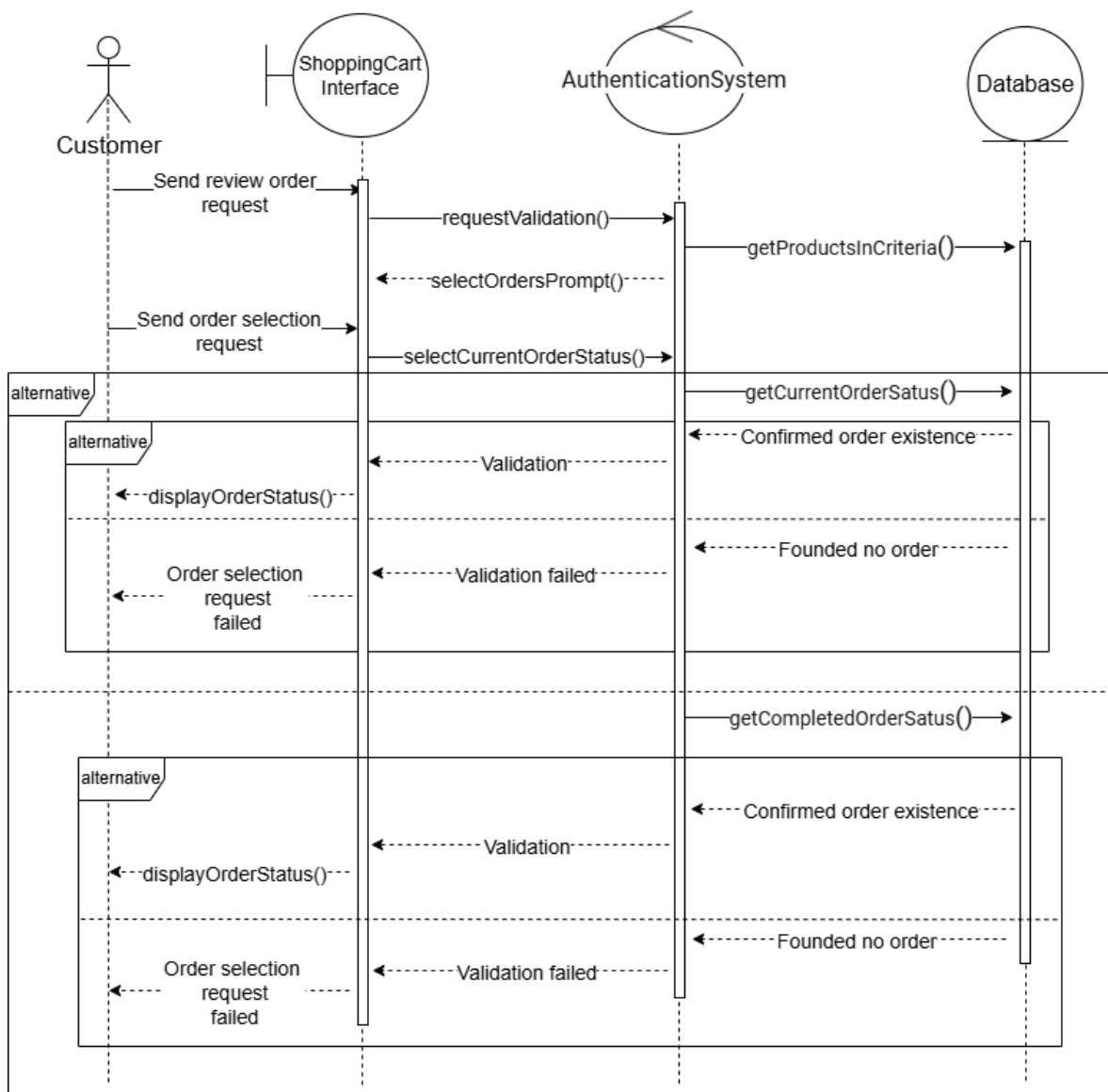
6.4 Update product sequence diagram



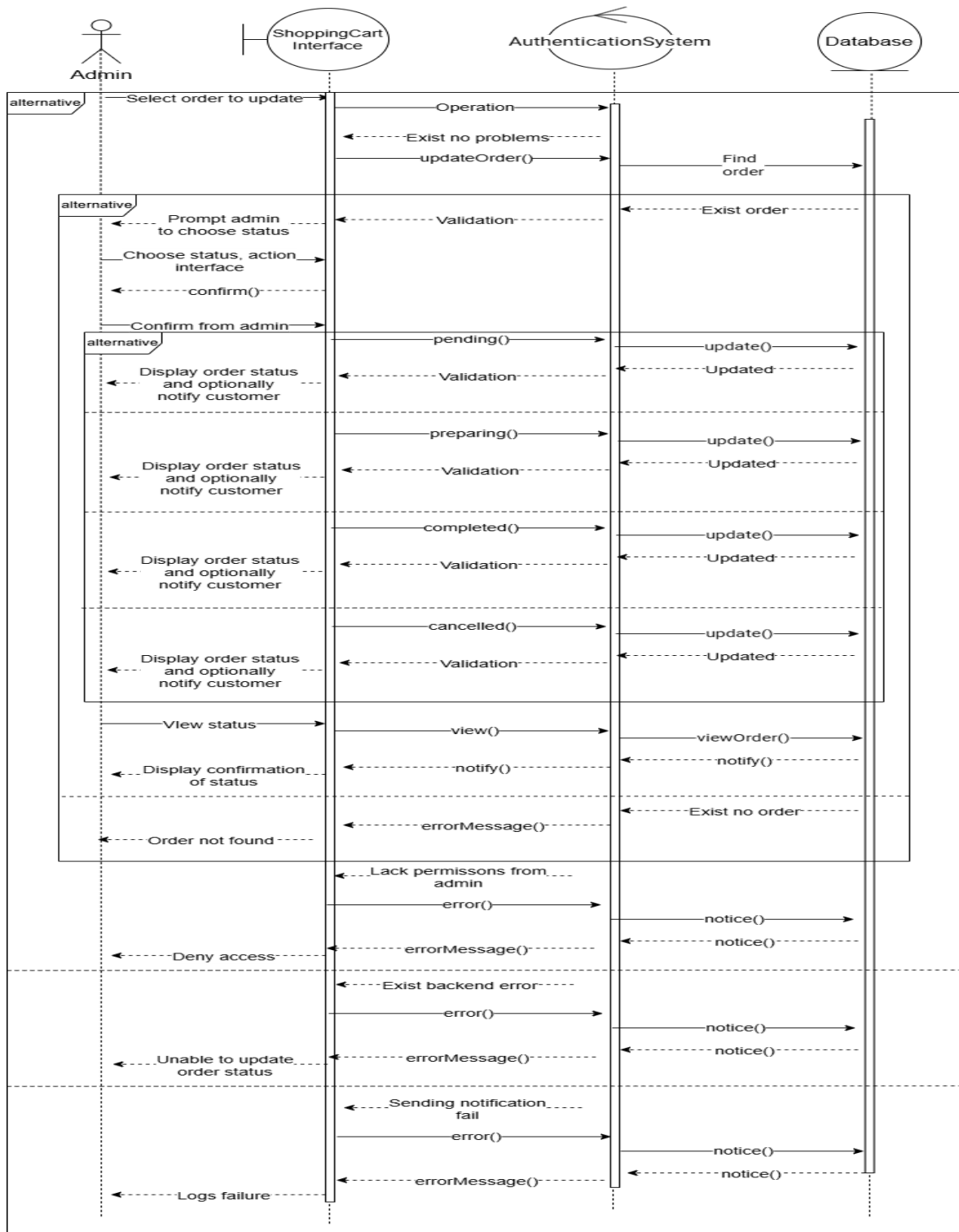
6.5 Delete product sequence diagram



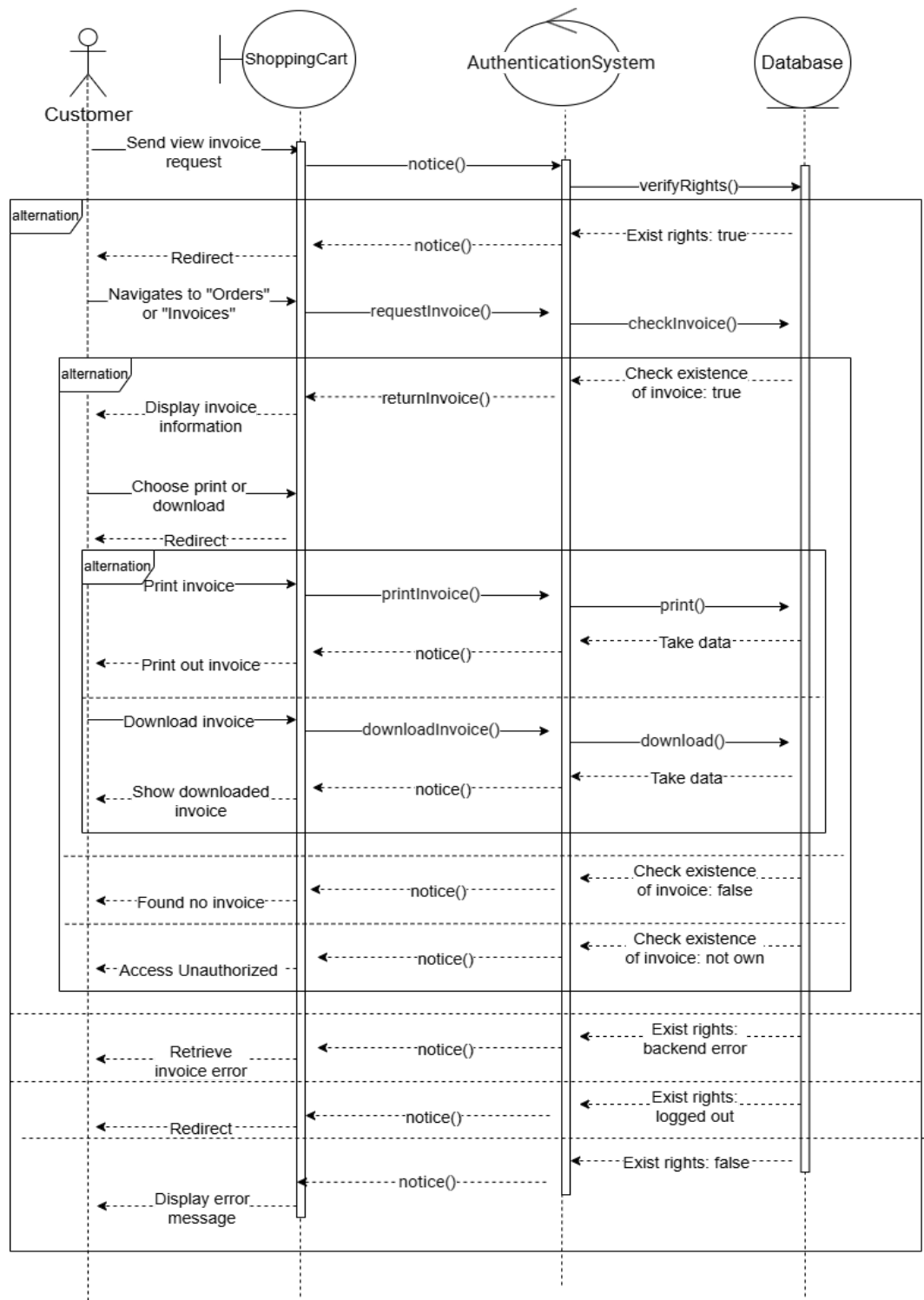
6.6 View order sequence diagram



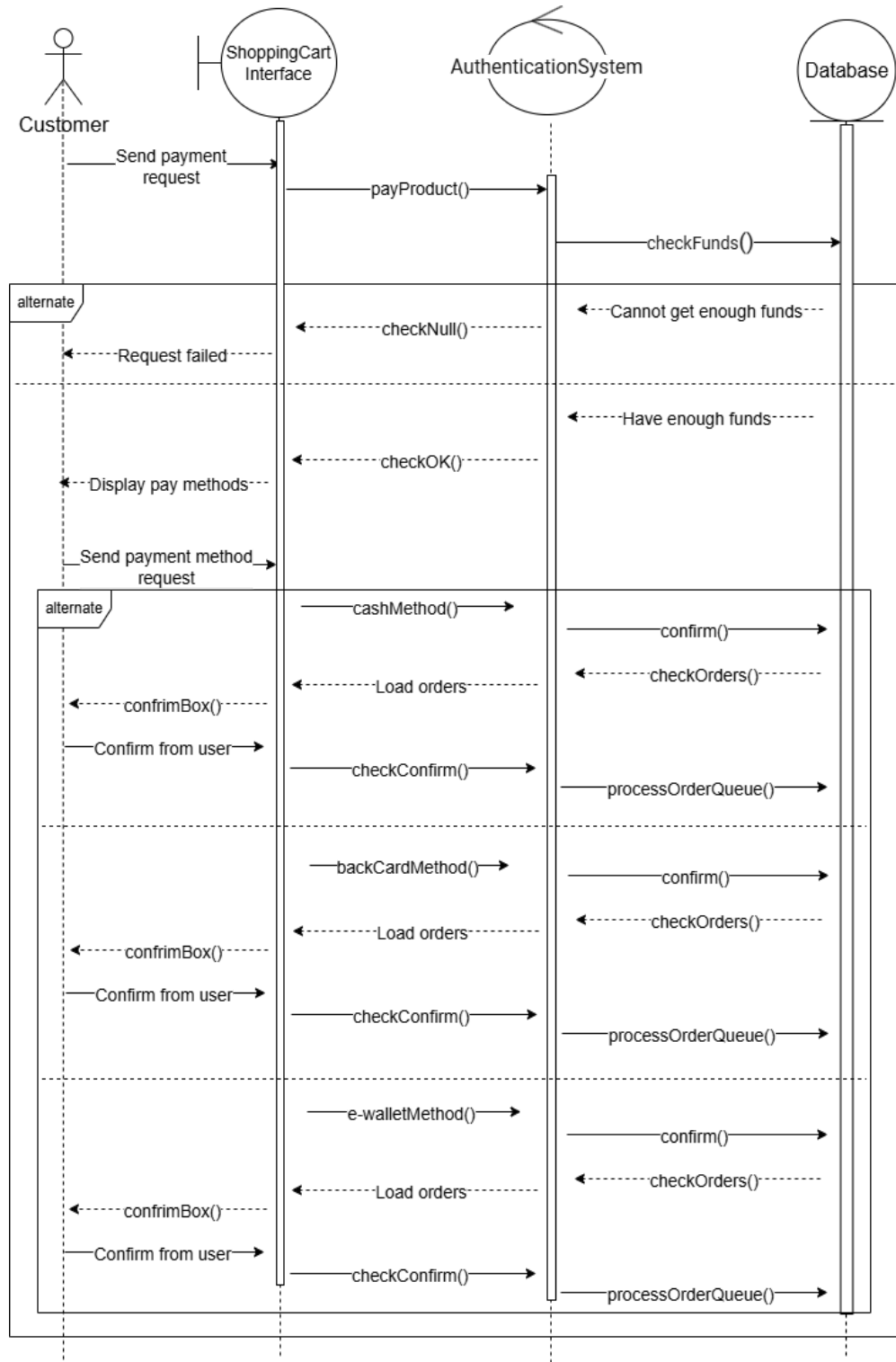
6.7 Update order status sequence diagram



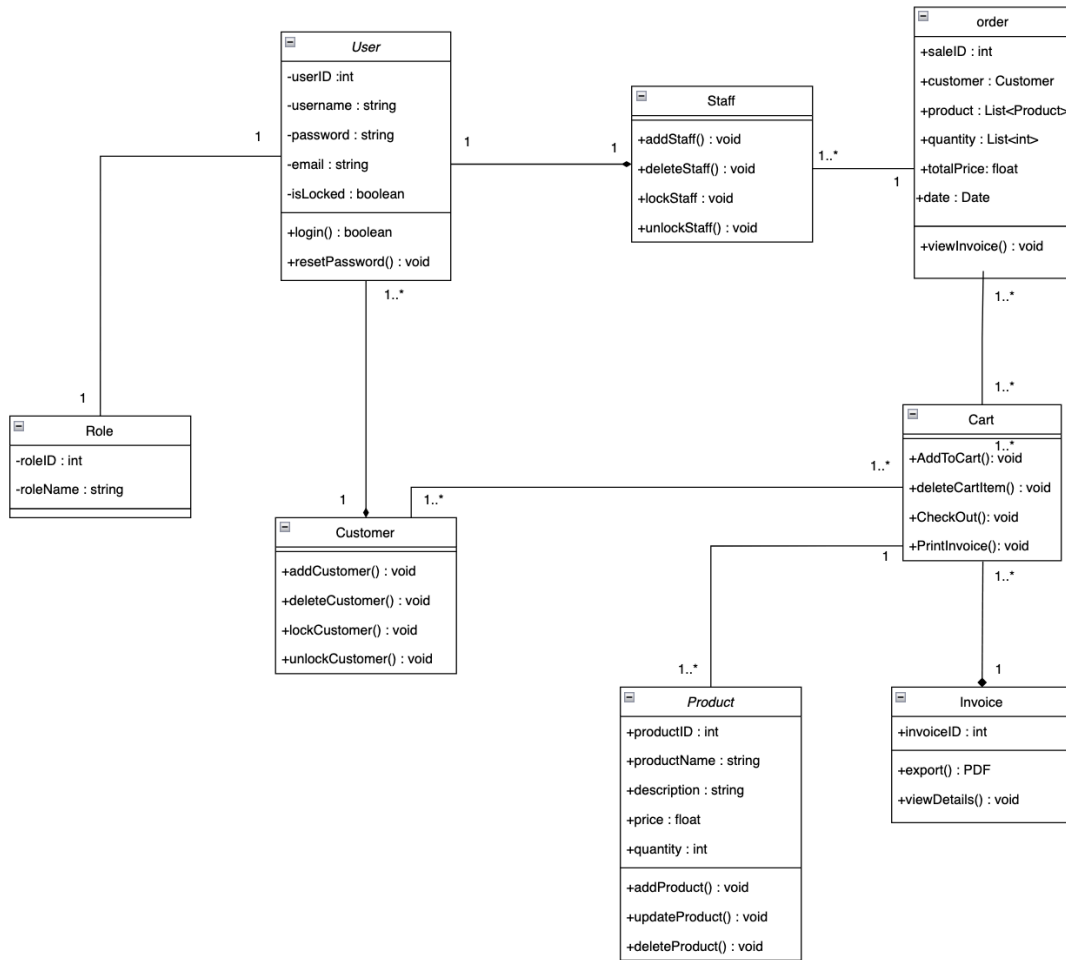
6.8 View and print invoice sequence diagram



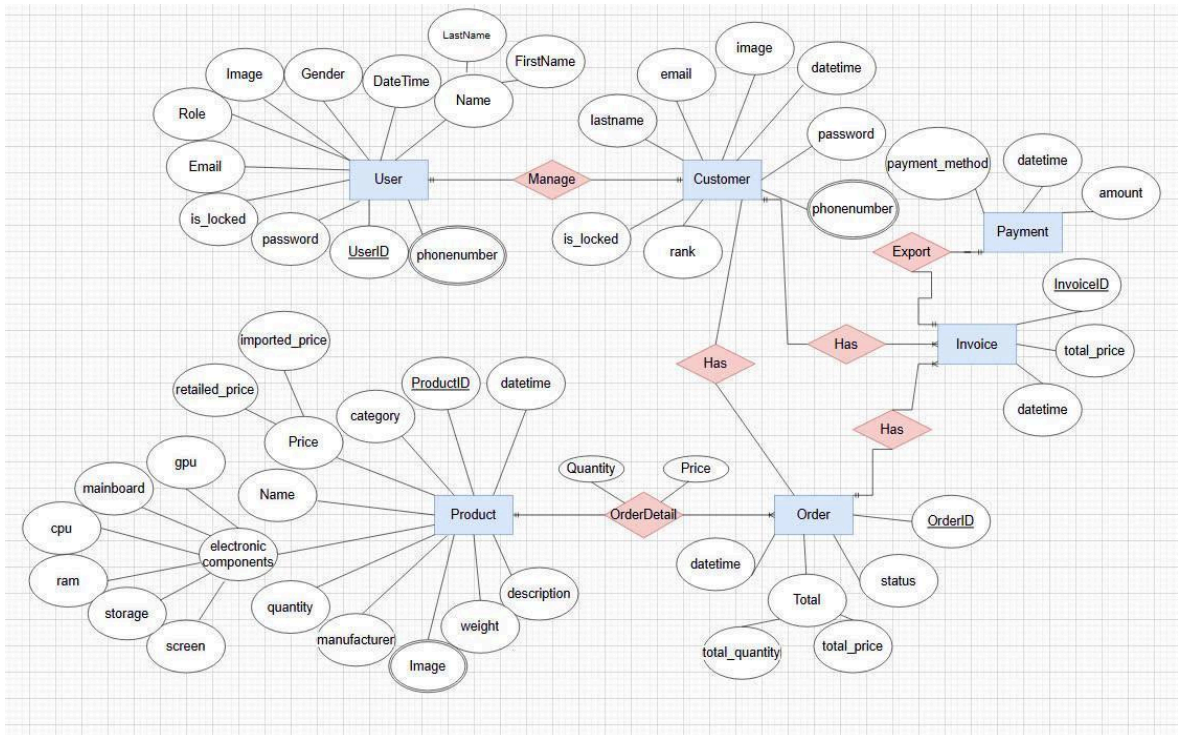
6.9 Checkout sequence diagram



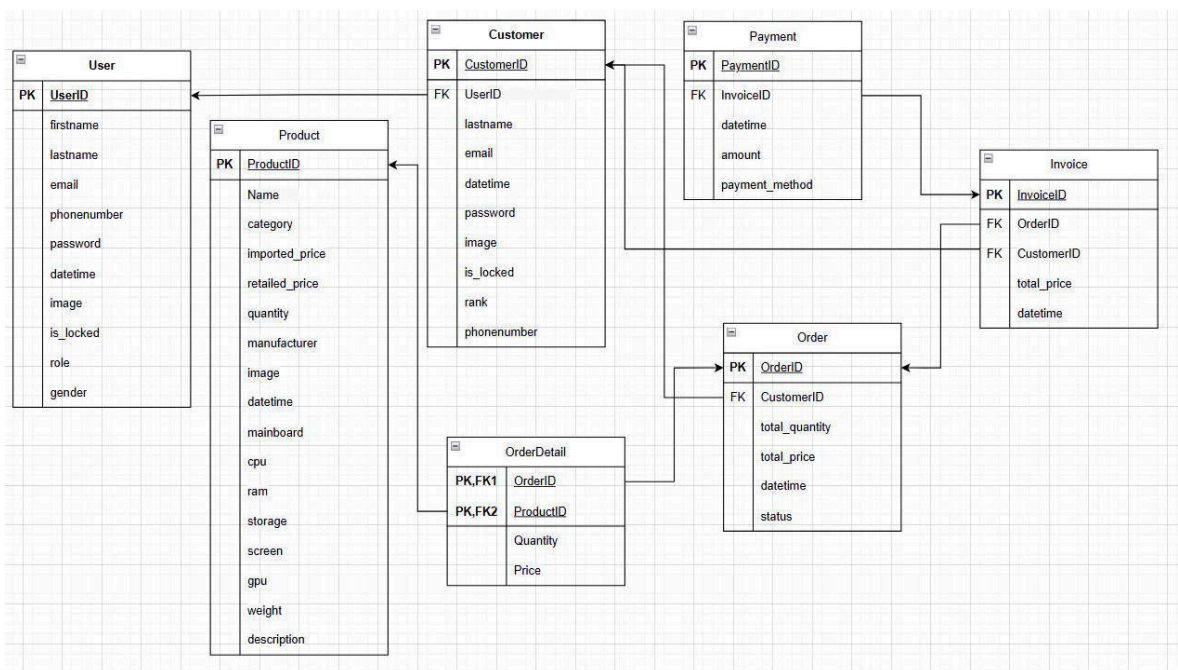
CHAPTER 7. CLASS DIAGRAM



CHAPTER 8. ENTITY RELATIONSHIP DIAGRAM



CHAPTER 9. RELATIONAL DATA MODEL



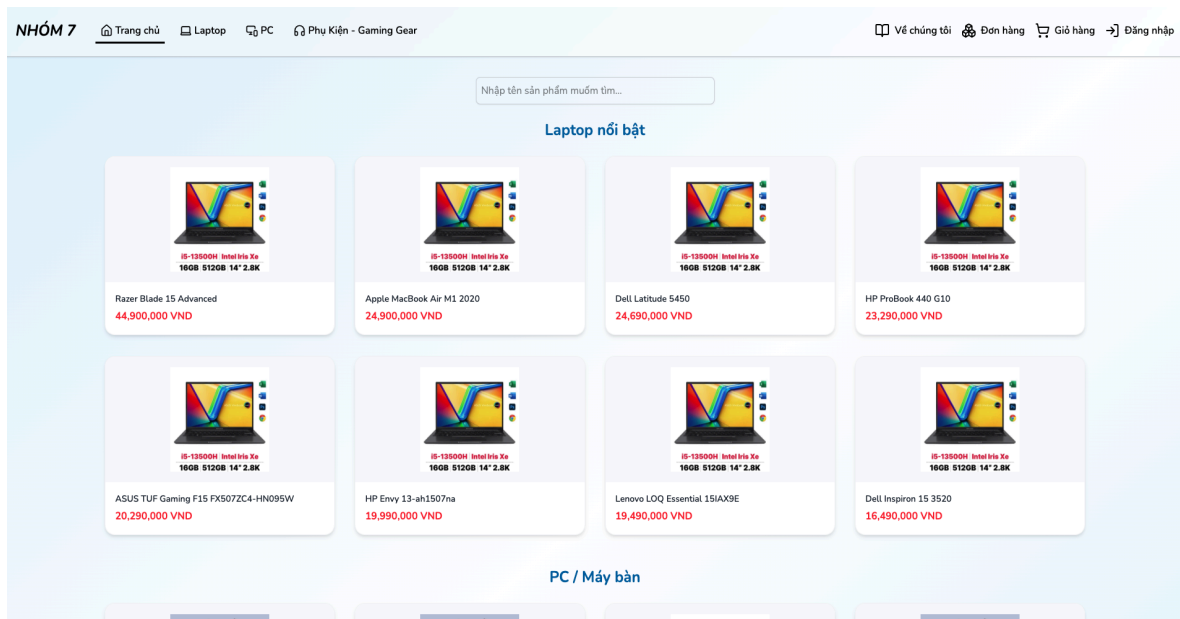
CHAPTER 10. APPLICATION

10.1 Github Repo

<https://github.com/DangKkhoa/computer-shop-management-react-complete.git>

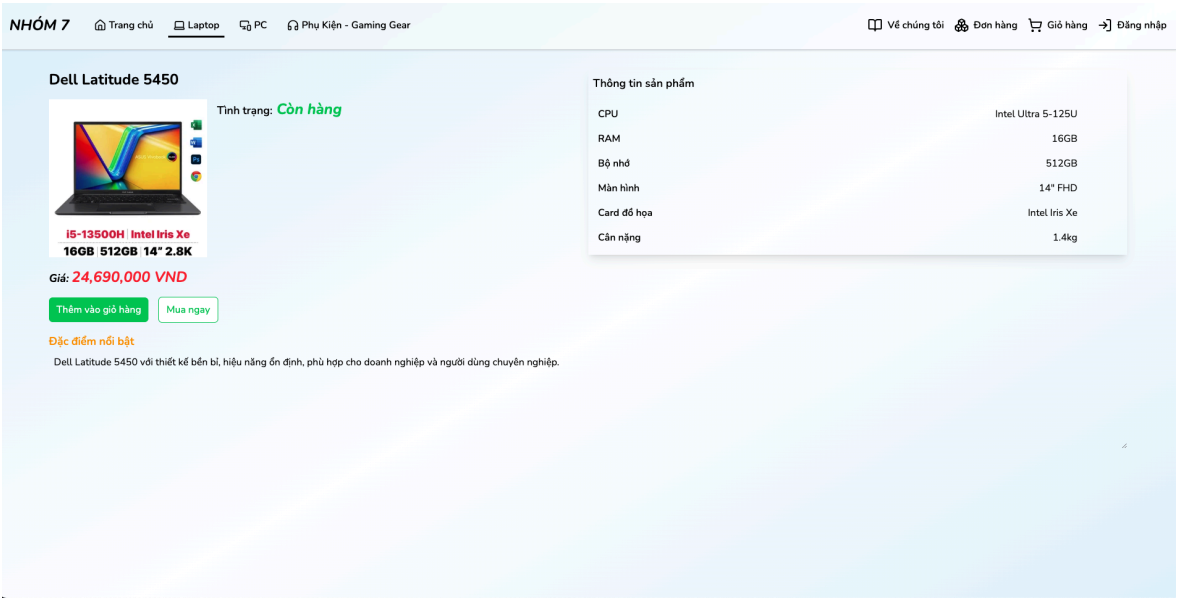
10.2 Application Images

10.2.1 Shop interface



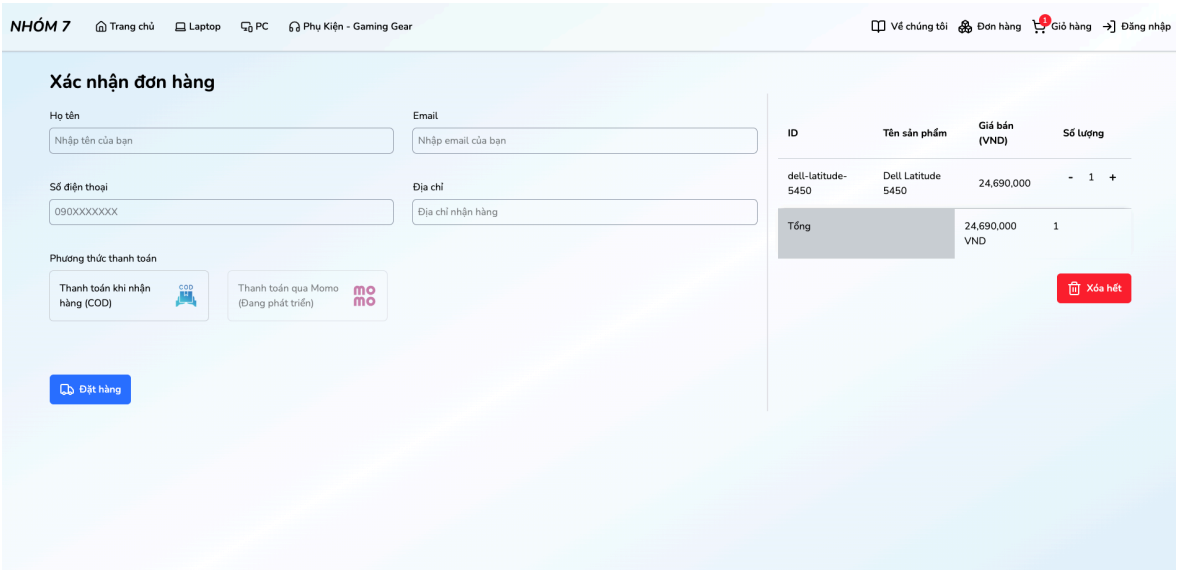
10.2.2 Product detail interface

Related use case: Manage product use case diagram



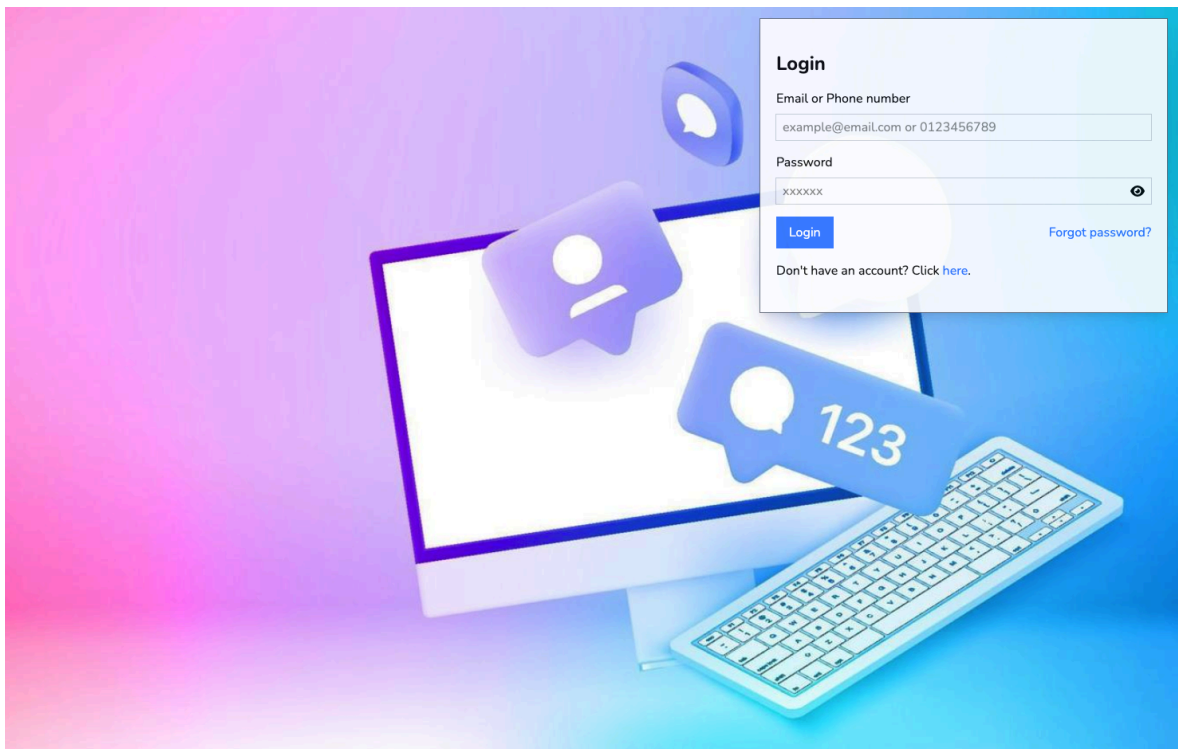
10.2.3 Cart interface

Related use case: Make payment use case diagram



10.2.4 Staff / Admin login interface

Related use case: Login use case diagram

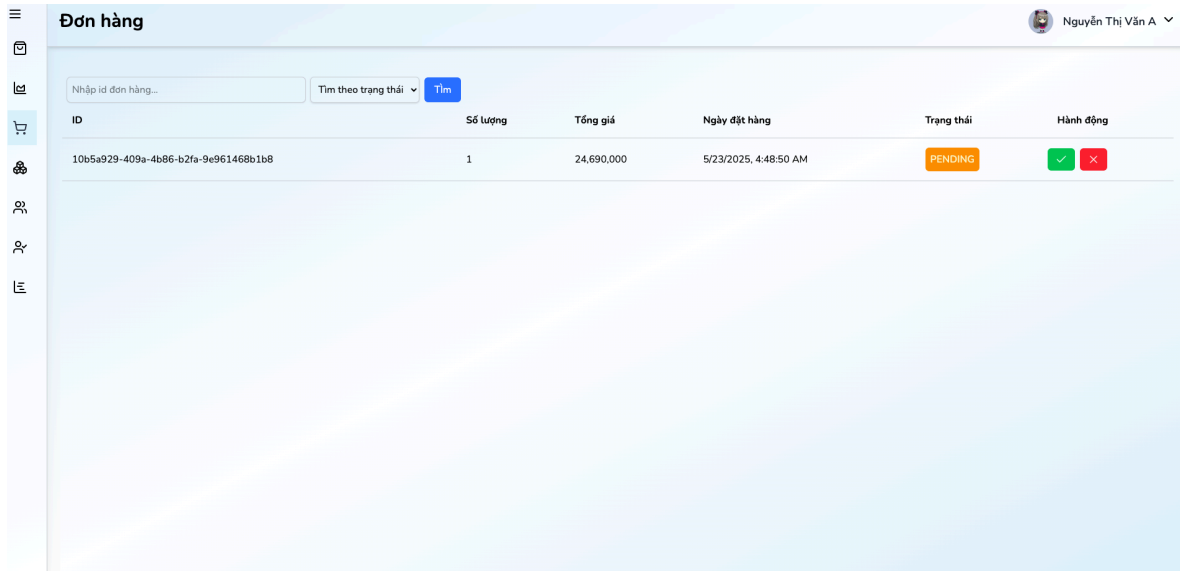


10.2.5 Admin dashboard interface



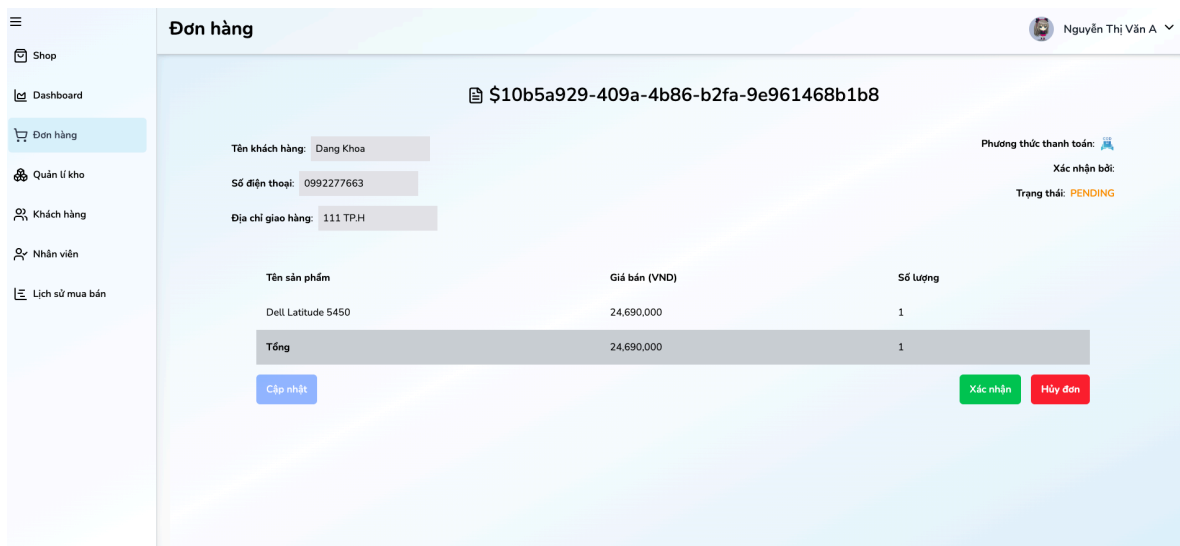
10.2.6 Order management interface

Related use case: Order management use case diagram



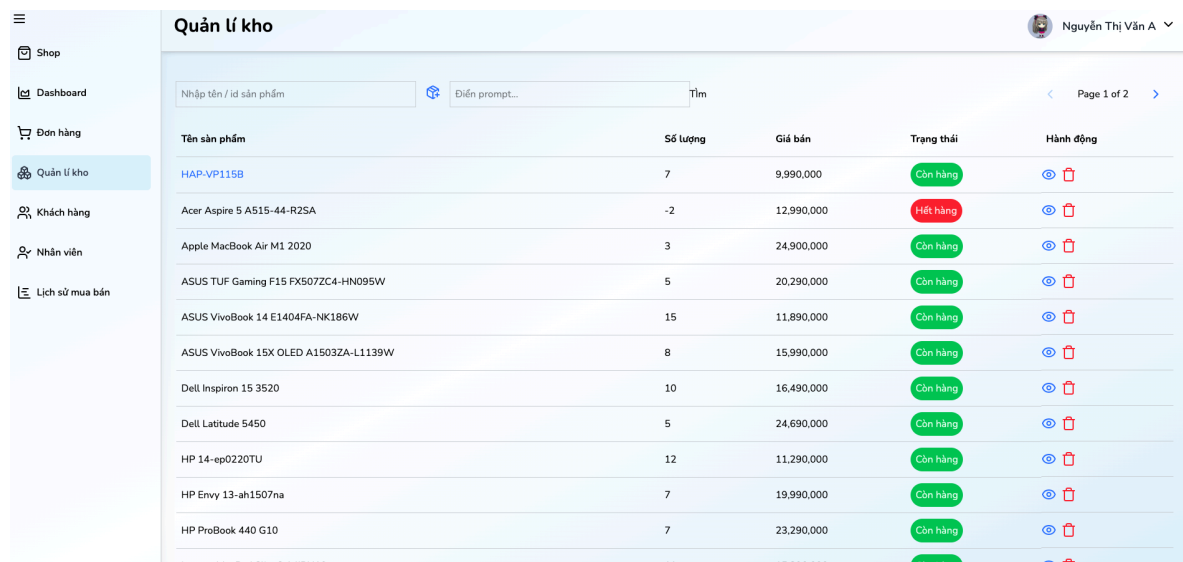
10.2.7 Order detail interface

Related use case: Order management use case diagram



10.2.8Inventory interface

Related use case: Product management use case diagram



10.2.9Product detail interface

Related use case: Product management use case diagram



CHAPTER 11. MEMBER EVALUATION

ID	NAME	TASKS	PERCENTAGE
521H0503	Truong Huynh Dang Khoa	1, 2, 3, 4.1, 4.2, 5, 10, 11	
519H0238	Nguyen Quoc Thinh	8, 9	
523H0082	Nguyen Viet Quan	6, 7	
523H0062	Nguyen Ngoc Nghia	5	

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2. <https://codelearn.io/sharing/sequence-diagram-trong-uml>
3. <https://viblo.asia/p/tim-hieu-ve-cach-thiet-ke-class-diagram-L4x5xLyY5BM>

English

1. <https://www.geeksforgeeks.org/top-8-software-development-models-used-in-industry/>
2. <https://www.ibm.com/docs/en/engineering-lifecycle-management-suite/doors-next/7.0.3?topic=cases-use-case-specification-outline>
3. <https://www.geeksforgeeks.org/use-case-diagram/>
4. <https://www.ibm.com/docs/en/rational-soft-arch/9.6.1?topic=diagrams-activity>
5. https://medium.com/@claire_logan/3-relational-data-model-examples-c9f70c61588c